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A review of multi-scale and multi-physics simulations of metal additive manufacturing processes with focus on modeling strategies

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ABSTRACT

Numerical simulations have recently shown their potential as a robust, cheap and reliable tool for predicting the quality of components produced by metal additive manufacturing (MAM) processes. Despite the advantages of the MAM processes over conventional manufacturing methods, there is still a lack of thorough understanding on how different defects can form and originate during MAM processing. In that respect, advanced numerical techniques recently developed have the ability to predict the occurrence of such defects. These techniques have paved the way to efficiently obtain the optimal processing window for targeted mechanical properties to satisfy the end-use design requirements. The aim of this review paper is hence to present and classify numerical simulations of MAM, not solely based on their length-scale as often seen, but also based on the involved physics, as well as the modeling strategies at both the meso-scale and part-scale. The paper is arranged in the following way: First, literature describing purely conduction-based heat transfer simulations at meso-scale are presented. This is followed by a review of fluid-based simulations of increasing complexity based on the treatment of free surface of the melt pool at meso-scale. Finally, contributions based on different part-scale modeling approaches with a focus on thermo-mechanical behavior are reviewed.

1. Introduction

Additive manufacturing (AM), according to American Society for Testing and Materials (ASTM), is one of the three major manufacturing techniques [1] (along with forming and subtractive manufacturing), and is used to produce metallic or non-metallic components from an input CAD file and via addition of feedstock material in a layer-by-layer manner. AM is divided into 7 different groups where the Laser Powder Bed Fusion (L-PBF), Directed Energy Deposition (DED), material extrusion and binder jetting processes fall into the branch of MAM.

MAM has a number of special characteristics such as low material waste, short total end-to-end production time, superior mechanical properties and above everything else, its capability of manufacturing complex geometries like lattice structures and topology-optimized geometries [2–4]. These have made MAM a superior process in many aspects compared to other conventional production techniques such as milling and casting. Due to these capabilities, many world-leading companies such as Boeing, Airbus and General Electric (GE), have become increasingly interested in MAM [5]. Furthermore, currently MAM is being implemented for the production of metallic parts with various applications such as medical, automotive, aerospace, energy, etc. [6–13]. Moreover, the MAM market has grown about 875% in the period of 2013–2018 [14] and the number of sold L-PBF machines has increased from 300 in 2013–1800 in 2017 [15], underlining the increasing interest of manufacturers in MAM as a production technique.

Despite the fact that MAM has the potential of becoming a superior production method, when compared to existing manufacturing techniques, there are still several possible forms of defects that can impair the overall quality of the final samples. Fig. 1 shows such different possible defects that can form in the MAM samples either during the process itself or its post-processing steps.

These defects, which cover a wide range in both shape, type and size, primarily originate from our limited knowledge about this very complex process. Usually an experimental trial-and-error procedure is carried out to find the optimal processing window to achieve a part with the desired quality. These experimental methods cover a large number of techniques. For instance, Light Optical Microscopy (LOM) and Scanning Electron Microscopy (SEM) are typically used for observing meso-scale features such as porosities and grain shapes [16–24], whereas the Fringe Projection (FP) method (a result of which is shown in Fig. 1(g)), is mainly used for determining and quantifying macro-scale deformations [25–27]. Other techniques, such as Electron Back-Scattered Diffraction

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(EBSD) or X-ray Computed Tomography (X-CT) are used for the visualization of grains and identification of porosities inside samples [16,20–23,28,29].

These experimental techniques are primarily conducted, after the part has been manufactured, and thus they are called *ex-situ* methods. Even though these *ex-situ* techniques can provide the manufacturers with a large amount of information at both micro- and macro-scales, they do not reveal the mechanisms underlying the formation of these defects. As a remedy, one can use in-situ monitoring techniques (such as infra-red thermal imaging and X-ray monitoring [30–32]) to acquire more detailed information regarding the formation and evolution of these defects, at the stage when the part is being manufactured.

Nevertheless, these online monitoring techniques need lengthy preparation and a laborious calibration. Alternatively, applying numerical models can provide a more cost-effective solution compared to in-situ monitoring. Furthermore, as opposed to the experimental techniques that are solely aimed and designed for only a specific subject, the numerical models are much more flexible and can be readily modified upon the application. Depending on their implemented physics, these models can simulate phenomena at different length-scales, e.g. dendritic growth [33], melt pool instabilities [34] at micro- and meso-level and stress formation and crack propagation [35] at macro-level (so-called part-scale), respectively.

Researchers have made numerous contributions in recent years in the field of MAM. This also goes for its modeling and simulation, where there has been a large number of contributions spanning from dendritic and grain growth models to pure thermal models, thermo-mechanical models, high-fidelity and finally reduced-order ones. In this respect, as the number of research output within MAM simulations, is steadily growing, many early-stage researchers and newcomers to the field might find it quite difficult to have a clear grasp of the current state-of-the-art and its relevant literature. In this way, review papers can be very helpful, as they collect and briefly describe hallmarks within a specific field.

There has already been a number of review papers within the field of MAM simulations. For instance, the review paper by King et al. [36] in 2014 from Lawrence Livermore National Laboratory (LLNL), introduced one of the earliest classification of MAM simulations and then divided the models into three classes of microstructural models, powder models (ALE3D code) and effective medium models (Diablo code). Their review paper, in essence, was mostly oriented around the numerical works done within LLNL and on the dross formation at overhang features. In another work, Korner and Markl [37] introduced different modeling techniques for the L-PBF and Electron beam Powder Bed Fusion (E-PBF) processes in 2016. They instead focused on the research outputs from FAU (Friedrich-Alexander Universitat) and confined their discussions mainly to powder-scale models. Meier et al. [38] in 2017 came up with a new classification of models, that is purely based on their length-scales. Meier et al. [38] accordingly introduced the three divisions of micro-, meso- and macro- (part) scales. This categorization based on length-scale, is currently the most widely-accepted classification in this field, thus in this work, we also adhere to this classification.

The mentioned review papers are becoming relatively old, especially since this specific field progresses very fast. Among more recent works, is the review paper by Bartlett and Li [39] in 2018, which is focused on the formation and evolution of residual stresses in MAM parts. The work reviews both experimental and numerical methods for stress analysis. Luo and Zhao [40] in 2018, reviewed the existing thermo-mechanical models based on finite element (FEM) for the L-PBF process, and then summarized the pros and cons of these models. Bertini et al. [41] in 2019, made a review of different modeling strategies for simulating the residual stresses for L-PBF. In a more recent work in 2019, Cook et al. [42], reviewed meso-scale computational fluid dynamics (CFD) models used for the L-PBF process. In their discussion part, they go through different simulation frameworks developed by well-known research groups in the field. Zhang et al. [43] reviewed the most recent progress in modeling of microstructural evolution for titanium alloys and at the micro-scale. Debroy's group has made two outstanding review papers in the field of MAM [44,45]. In their later work, Wei et al. [45] go through most of the existing models in the field of MAM up until 2020. In their work, they start from the very basic modeling tools, such as moving point heat sources, all the way to complex CFD simulations implemented for MAM modeling. However, they barely touch upon residual stress modeling, and by that they nearly do not enter the part-scale simulations.

In this review paper, rather than going into details of a certain type of model, or going through nearly all types of models without any clear classification, we aim to give an overview of possible modeling strategies at both meso-scale and part-scale simulations for MAM. In this respect, the meso-scale and part-scale simulations of MAM are first enumerated and are then classified based on their modeling strategies. The rest of this manuscript is organized as follows: in Section 2, DED and

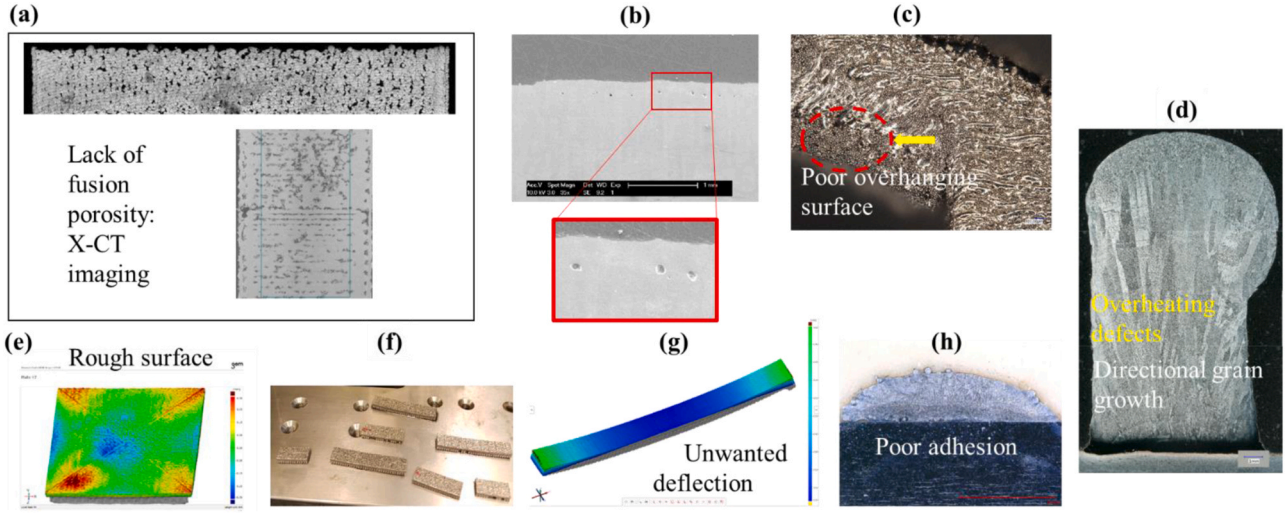


Fig. 1. An overview of a number of possible types of defects that can form during the MAM process or its subsequent post-processing steps. (a) and (b) show lack-of-fusion and keyhole porosity via LOM, X-CT and SEM analysis. (c) shows poor overhang structure. (d) shows the impact of overheating during the multilayer DED process and also directional grain growth. (e) shows rough surface of a plate. (f) and (g) show the crack formation and unwanted deflection after cutting the samples from the base plate. (h) shows poor adhesion of a track made by DED.

L-PBF processes are first briefly introduced and then the challenges arising from the multi-physics and multi-scale nature of the process are described. This section also introduces the classification of the simulations based on their modeling strategy. In Sections 3 and 4, the meso-scale models for MAM are reviewed and discussed, where in the former pure conduction models and their possible modeling strategies are considered and in the latter, different classes of CFD models are discussed. In Section 5, part-scale models and different multi-scaling laws used for such simulations are elaborated and discussed. The final section contains a conclusion on the described models. Furthermore, at the end of each sub-section, a discussion of the pros and cons of the implemented modeling technique is given as well.

2. MAM processes and their underlying physics

2.1. The L-PBF and DED processes

Fig. 2(a)-(c) show different steps of the L-PBF process. According to Fig. 2(a) in the first step, the powder table moves an increment of approximately one layer of powder upwards, while the build table moves the same distance but downwards. Then, the coating tool adds a layer of powder particles on the build platform, see Fig. 2(b), and in the last step, the laser starts fusing discrete powder particles together, Fig. 2 (c). These steps repeat until the part is manufactured, with a time gap denoted the cooling time at the end of each cycle.

The DED process, however, is more continuous than the L-PBF process. In DED, a continuous flow of material, either in the form of powder streams or wire, is delivered to the location which is being irradiated by the laser beam. These powder particles are carried by the gas flow and the irradiation zone itself is constantly protected by a shielding gas flow during the course of the process, see Fig. 3. When the molten powder particles collide with the base plate, they form a track of metal, at the wake of the laser beam.

2.2. Physics and scales in MAM

MAM is inherently a complex process due to two main reasons. First, its multi-physics nature as it involves complex interacting physical phenomena and second, due to its multi-scale characteristics, meaning that these different phenomena occur at different length scales. Furthermore, what makes modeling MAM even more challenging, is the fact that some of these physics occur in a very short time period in the order of microseconds. Fig. 4 shows the temperature contour and the velocity field vector from a meso-scale multi-physics simulation.

According to Fig. 4, the amount of energy is sufficiently high to melt the powder particles, despite the fact that a larger portion of the delivered energy is reflected rather than being absorbed by the powder particles. Once the melt pool is formed, interfacial forces will gradually appear on the surface of the melt pool and act as the main drivers for the fluid flow in the liquid domain. The origin of two of these forces, i.e. surface tension effects and thermo-capillarity (the Marangoni effect), are directly related to surface tension. The latter, at least for a typical alloy without high concentration of surface active elements, leads to an outward flow within the melt pool and depends on local temperature gradients. Meanwhile, the former acts perpendicularly to the melt pool surface and increases with declining temperature. Similarly to the surface tension force, the third interfacial force is the recoil pressure that acts normal to the free surface and pushes the liquid downwards. This force, under some special conditions, leads to the formation of keyhole depression zones that might lead to porosities as well.

Another major phenomenon in MAM is the formation and evolution of residual stresses. During the period in which the laser irradiates the metal, material melting and solidification cause thermal strains. As soon as the laser leaves this location, solidification starts. During this very short time period when the material is near melting temperature, the yield stress is very low, and since the melt pool in front of it expands, it yields in compression. Then, as the material behind it continues to cool, it would experience a tensile stress and undergo yielding. Eventually,

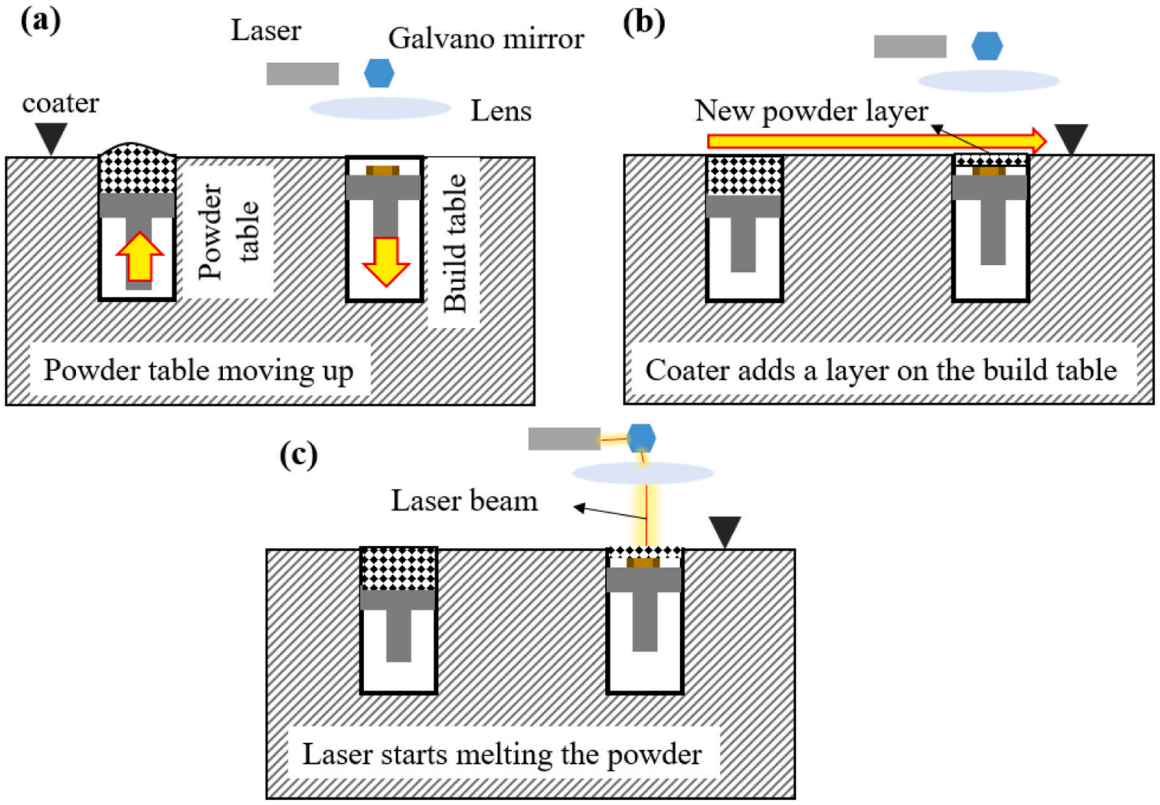


Fig. 2. (a)-(c) schematics showing different steps of the L-PBF process.

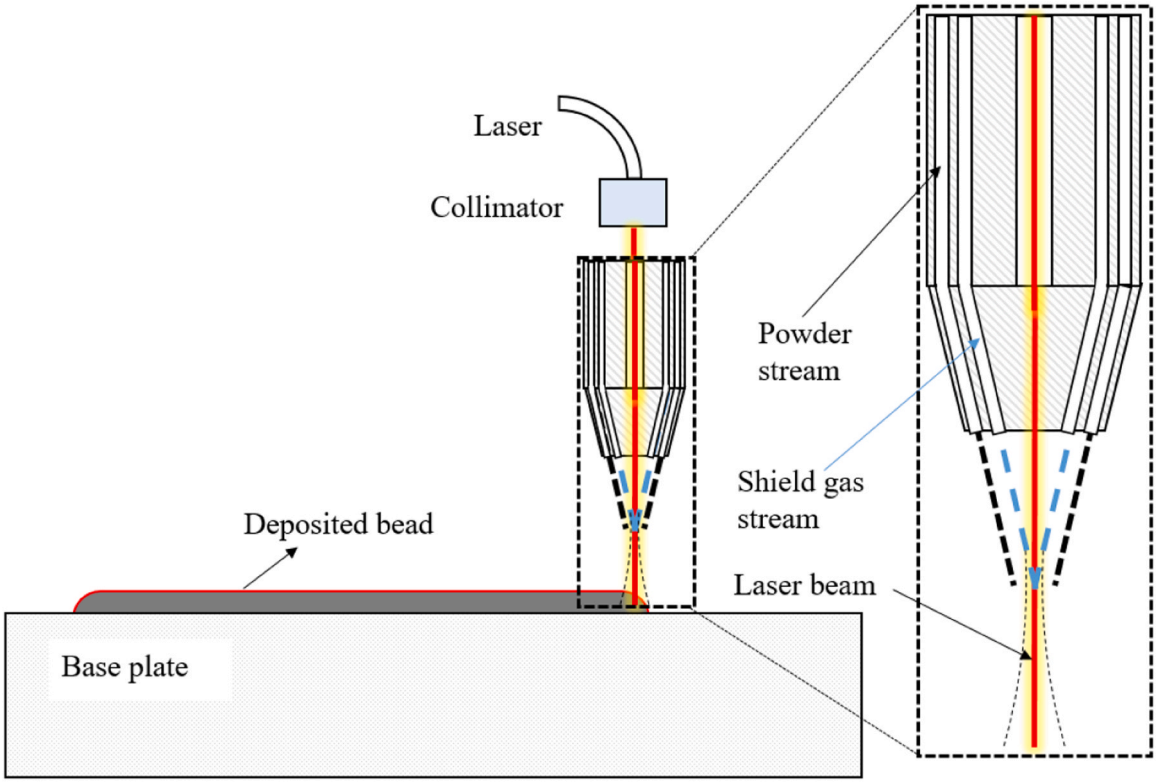


Fig. 3. A schematic view of the DED process.

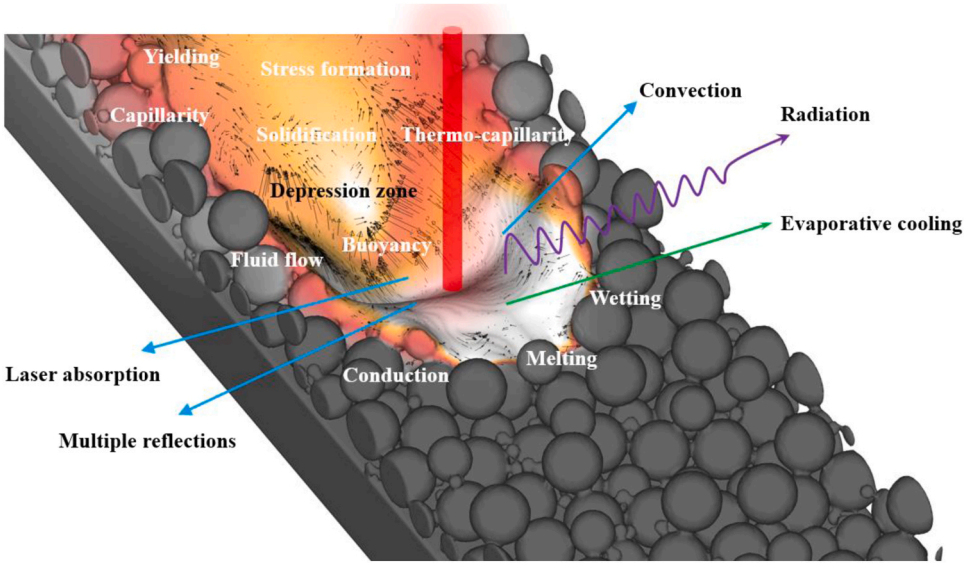


Fig. 4. Different physical phenomena occurring during MAM processes. This figure is taken from an unpublished simulation of the L-PBF process made by the authors.

the solidified track would end up in a tensile stress state while the already-built tracks nearby experience compressive stress. These residual stresses play an important role when the sample is cut and detached from the base plate as they could lead to undesired deflections, thus compromising the part quality. This, nevertheless, can be avoided if a stress relief is carried out right after the MAM process [46] whilst the parts are still attached to the build plate.

Currently, it is impossible to simulate all relevant physical phenomena including both the fluid flow and stress evolution at part-scale level, given today's computing capabilities and existing MAM modeling approaches. However, in recent years, there has been some contributions to couple an initial CFD simulation to a subsequent solid mechanical model, that receives its thermal input from the CFD model [47,48]. However, due to very high computational requirements, these simulations were only carried out at meso-scale or melt pool level including one or two tracks only. Two examples of this sort of modeling method are shown in Fig. 5, for E-PBF and DED.

When setting up a proper model, one must first specify the required outputs and then the length-scale of the model. For instance, if the focus is on predicting the keyhole porosity formation, a CFD model at the

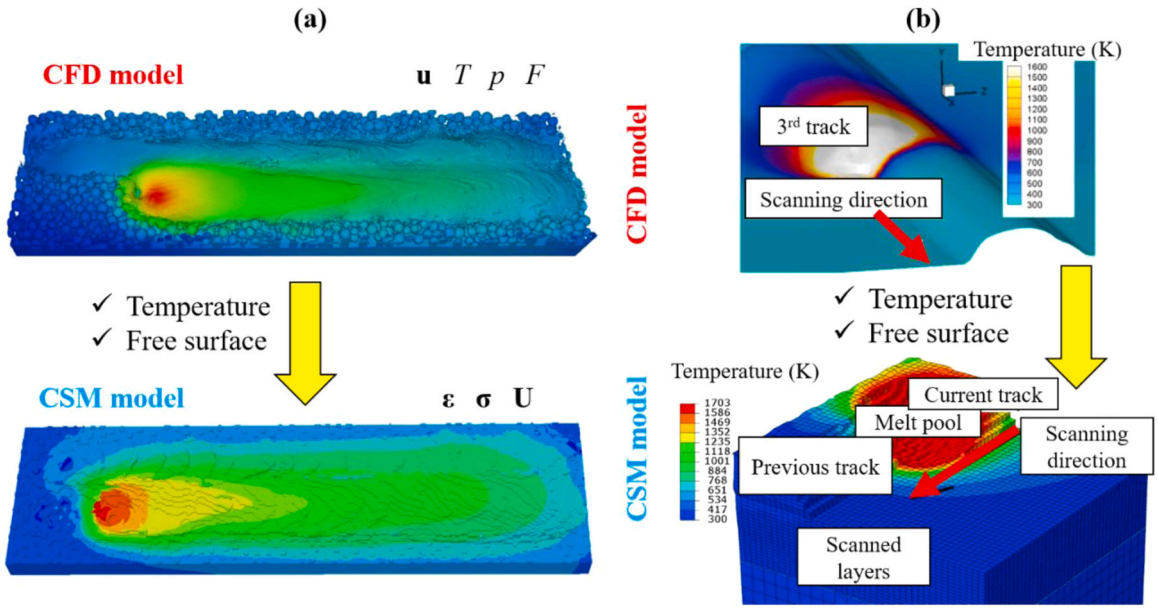


Fig. 5. Two examples showing the coupling of a CFD model that provides input to a subsequent mechanical (CSM) model for (a) E-PBF [47] and (b) DED [48].

meso-scale would be the solution; and if macroscopic deformation is the objective, a thermo-mechanical or mechanical model at part-scale is needed. On the other hand, if the impact of detailed scanning strategy on residual stresses is the target, a thermo-mechanical model at meso-scale that can resolve the movement of the laser beam needs to be developed. Fig. 6 shows some of the most common modeling strategies at both meso- and part-scales.

According to Fig. 6, heat transfer constitutes the core in any MAM simulation and it can be inferred that this physics is the driver of other physical phenomena such as fluid flow, stress formation, etc. For instance, by coupling the heat transfer to fluid dynamics at meso-scale, one can determine detailed thermo-fluid behavior within the melt pool.

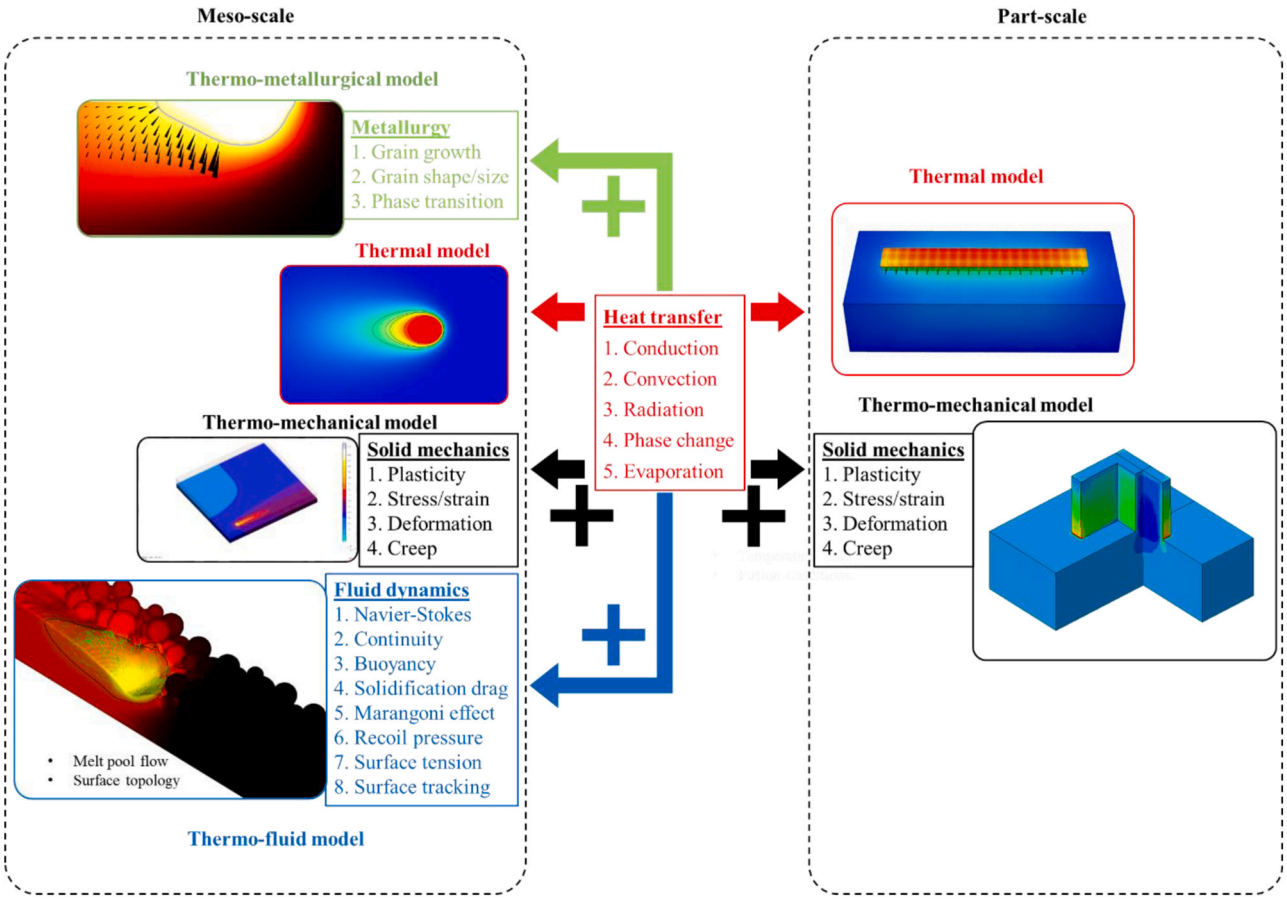


Fig. 6. An overview of current possible modeling strategies for MAM at both meso- and part-scale. colors for the internal borders show a specific modeling method.

This model can also be implemented for the prediction of porosity formation during the process, which will be discussed later in Section 4 of the paper. Likewise, by coupling heat transfer to a mechanical solver, one can simulate the thermo-mechanical conditions during the process. Furthermore, some of these modeling strategies can be divided into two groups of meso- and part-scale simulations, as shown in Fig. 6.

3. Meso-scale models: conduction-based simulations

3.1. Pure thermal models

A large portion of the scientific contributions related to MAM simulations is based on pure conduction models and largely up until 2018. In these models, the transient temperature field during the process is predicted via solving the partial differential equation arising from the energy balance, see Table 1, however, with the velocity vector put to zero. A_{accomd} is the accommodation coefficient in Table 1 and p_{atm} (Pa) is the reference pressure.

Most of these thermal models are developed based on FEM, although a handful of them are also FVM-based simulations, see Fig. 7(a) and (b). In the conduction models for the L-PBF process, the powder layer is assumed to be a continuum with averaged thermo-physical properties e.g. conductivity, specific heat capacity and density.

In this class of models, which corresponds very much to the thermal models used in computational welding mechanics, the laser-material interaction is simulated via either a moving heat flux or volumetric heat source with a planar Gaussian profile. An example of such a simulation is the work done by Li and Gu [50], who modeled the thermal conditions during L-PBF for commercially available pure titanium. In their work, a moving Gaussian heat flux was used to model the laser interaction with the powder bed. They found that an increase in the laser power would lead to a significant increase in temperature gradients, while a decrease in the scanning speed causes a slight change in thermal gradients. Liu, Zhang and Peng [51] developed a FEM thermal model to study the thermal-cycling due to repeated heating and cooling as subsequent layers are scanned for L-PBF of 316-L stainless steel. They reported bigger melt pools at subsequent tracks due to the residual heat effect. This aligned well with findings of Roberts et al. [52] who had previously shown, via FEM modeling of multi-track-multi-layer L-PBF that, as a result of residual heat effect, the temperature in the previously-deposited tracks steadily rises while new tracks/layers are being printed. Zhang et al. [53] developed an FEM-based thermal model to simulate the L-PBF process of W-Ni-Fe powder. They used the model to obtain optimal process parameters for printing. Although by 2015, most of the simulations were 3D, Criales, Arisoy and Ozel [54] did a 2D thermal model of the L-PBF for Inconel 625 (IN625). They carried out a sensitivity analysis to understand the influence of variations in different process parameters and material properties on the melt pool's temperature. They found that among 8 parameters (spot size diameter, conductivity, latent heat, power, scanning speed, specific heat, packing density and powder reflectivity), powder packing density can significantly affect the melt pool's temperature [54]. Shi et al. [55] and Antony et al. [56] developed heat conduction models based on FEM with a Gaussian heat flux describing the heat input for L-PBF of IN718 and 316-L, respectively and used these to identify optimal printing conditions.

Table 1
Partial differential equation of energy balance and its related boundary conditions and phase change equation.

Heat transfer	$\frac{\partial}{\partial t}(\rho \quad h) + \frac{\partial}{\partial x_j}(\rho \quad hu_j) = \frac{\partial}{\partial x_j}\left[k_{bulk}\left(\frac{\partial T}{\partial x_j}\right)\right] + \dot{Q}_V^{''}$
Phase change (linear)	$f_{liq} = \frac{T - T_{sol}}{T_{liq} - T_{sol}}, \quad T \in [T_{sol}, T_{liq}]$
Thermal boundary conditions	$-k\frac{\partial T}{\partial \boldsymbol{n}} = q_{rad}^{''} + q_{amb}^{''} + q_{evap}^{''}$
$q_{rad}^{''}$ (radiation)	$\varepsilon\eta\left(T^4 - T_{surr}^4\right)$
$q_{amb}^{''}$ (convection)	$h(T - T_{surr})$
$q_{evap}^{''}$ (evaporation)	$\frac{A_{accomd}}{\sqrt{2\pi R_v T}}\left[p_{atm}\exp[f_0]\left(\frac{\Delta H_{lv}}{R_v T_{boil}}\left[1 - \frac{T_{boil}}{T}\right]\right)\right]$

3.1.1. Capabilities and limitations of conduction models

The pure heat conduction models imply very coarse assumptions with respect to the real MAM process, because they do not take the fluid dynamics into account, which is a major mechanism of heat transfer in the melt pool. In this respect, this type of models can mainly be used for optimization studies in which mere sensitives are needed instead of absolute values, and these models often need to be re-calibrated for new process parameters, effectively making them applicable to a limited process window, only. Table 2 lists some of the major contributions within pure thermal models of MAM, together with their highlights and modeling techniques.

As the laser rays can penetrate into the powder bed and this mainly is due to multiple reflections, one can model the laser-material interaction using a moving volumetric heat source as well. For instance, Foroozmehr et al. [60] came up with a cylindrical volumetric heat source which has a Gaussian planar profile in essence. They changed the depth of the heat source as a way to calibrate their thermal model. Yin et al. [63] adopted an exponential-decaying function for their heat source distribution in the vertical direction, while having a Gaussian profile in the horizontal plane. Nevertheless, Tran and Lo [61] did not model their heat source based on pre-existing functions and they ran a first-step Monte-Carlo ray-tracing method in a packed powder domain. They found the absorption profile from this ray-tracing model and later implemented that in the volumetric heat source in their FEM thermal model. There are also two other well-known heat sources used for MAM simulations, Goldak's double-ellipsoid [64] and Gusarov's heat source [65]. The former is essentially inherited from the welding community. Table 3 lists some of the heat sources that are typically used in MAM simulations.

Bruno-Rosso, Demir and Previtali [62] developed an FEM-based thermal model and used Goldak's double-ellipsoid heat source for validation and then Gusarov's heat source for parametric study. Vastola et al. [67] used the heat source given in Table 3 and used that for thermal modeling of the keyhole-mode melt pool during L-PBF, even though their model was a pure conduction model without any surface-tracking technique.

3.2. Thermo-metallurgical models

Pure conduction models according to Kamara et al. [70], overestimate the melt pool temperature, as they neglect the convective heat transfer from the fluid dynamics. Kamara et al. [70] suggest that, one can use a non-isotropic effective liquid conductivity in order to avoid the overestimation of peak temperatures, and more importantly, to compensate for their low fidelity. Such models with an effective liquid conductivity will be introduced later in this paper.

Despite these limitations, pure conduction models are robust and computationally cheap [71]. In this respect, these pure conduction models can be further exploited by coupling to other physics such as metallurgical or mechanical models. Even in some cases, one can use these models to perform an optimization to obtain the optimal processing condition to reduce final deformation, similar to the work by Mohanty and Hattel [72], based on a reduced-order pure conductive model. In the thermo-metallurgical models, the two independent solidification parameters, namely the solidification cooling rate and temperature gradient, are first calculated at the end and at the onset of solidification, respectively, see Fig. 8(a). When these parameters are determined, the solidification growth speed R ($m{\cdot}s^{-1}$) along with the morphology factor F ($K{\cdot}s{\cdot}m^{-2}$) can be obtained as well. According to Fig. 8(b), a higher morphology factor F, leads to columnar grain structure, while a low F value promotes equiaxed grains.

Thermal models for this class of simulations are normally either

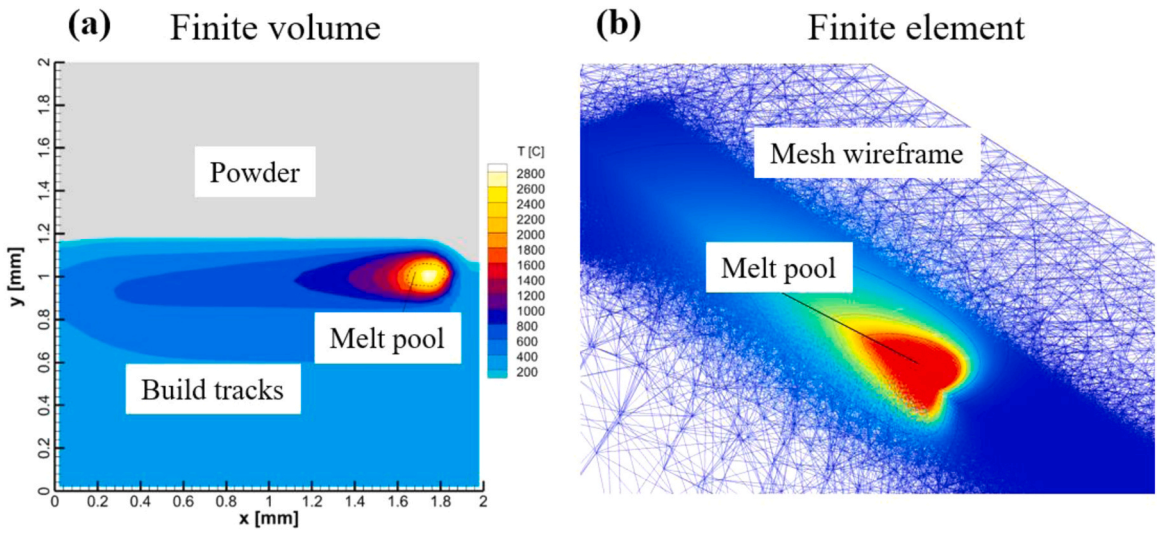


Fig. 7. Results of thermal simulation for the L-PBF process using (a) FVM [49] and (b) FEM frameworks.

Table 2
A brief overview of some of the most important pure conduction models used for thermal simulations.

Topic and highlights	Authors	Coupling	Software	Reference
3D thermal model of the L-PBF process	Yin et al. 2012	None	ANSYS	[57]
2D thermal model of L-PBF + comprehensive parametric study on the impact of 8 process condition on the thermal field within the melt pool	Criales, Arısoy, and Özel 2016	None	In-house code with Matlab	[54]
3D thermal model of L-PBF of W-Ni-Fe	D. Q. Zhang et al. 2010	None	ANSYS	[53]
3D thermal model of L-PBF of commercially pure titanium	Yali Li and Gu 2014	None	ANSYS	[50]
3D thermal modeling of L-PBF of 316-L stainless steel + modeling of the thermal cycling effect	Y. Liu, Zhang, and Pang 2018	None	ANSYS	[51]
Investigation of the residual heat propagation in multilayer L-PBF of Ti6Al4V	Roberts et al. 2009	None	ANSYS	[52]
Finding the optimal process parameters for achieving a dense morphology for L-PBF of IN718	Shi et al. 2016	None	ANSYS	[55]
Finding optimal process parameters for manufacturing defect-free tracks made of 316-L stainless steel	Antony, Arivazhagan, and Senthilkumaran 2014	None	ANSYS	[56]
3D thermal modeling of L-PBF of Al-6061 + considering the evaporation and shrinkage	Loh et al. 2015	None	COMSOL Multi-physics	[58]
3D thermal modeling of L-PBF of Al-6061 + vapor/liquid/solid phase indicator	Yingli Li et al. 2017	None	–	[59]
3D thermal modeling of L-PBF of 316-L stainless steel via implementing the optical penetration depth method	Foroozmehr et al. 2016	None	ANSYS	[60]
3D thermal modeling of L-PBF of 316-L stainless steel + developing a volumetric heat source based on a Monte-Carlo ray-tracing method	Tran and Lo 2018	None	COMSOL Multi-physics	[61]
3D thermal modeling of multi-track L-PBF process + validation with high-speed imaging	Bruna-Rosso, Demir, and Previtali 2018	None	In-house code	[62]
3D thermal modeling using a volumetric heat source with an exponential decay distribution along the depth of the powder layer	Yin et al. 2016	None	ANSYS	[63]

analytical or based on FEM or FVM. For instance, Bontha et al. [76] used Rosenthal's analytical solution for a moving point source to simulate thermal conditions during MAM for Ti6Al4V. In their work, it was concluded that an increase in the input laser power or a decrease in the scanning speed can both lead to coarser grain sizes and at the same time promote equiaxed grain formation, see Fig. 8(d), since the lower the G/R ratio, the higher will be the chance of nucleation of equiaxed grains. Plotkowski et al. [78] also used an analytical solution to a moving ring-shaped heat source in a semi-infinite domain to calculate the temperature field. In their study, they managed to study the metallurgical evolution of an entire layer. They specifically noted that a higher contact time of 0.25 s can remarkably affect the grain morphology and makes it predominantly equiaxed, as opposed to a spot time of 0.10 s, which would lead to a noticeable columnar grain.

Raghavan et al. [74] developed an FEM-based thermal model based on Truchas code [79] and coupled it to a metallurgical model for E-PBF of IN718. They showed that at the onset of solidification, grain morphology is mainly columnar, while at the end, the grain morphology transitions into equiaxed, see Fig. 8(c). This is also clearly demonstrated in the simulated grain morphology for laser welding in Fig. 8(c) and can be seen that the grain morphology is columnar on the edges of the melt pool and equiaxed at the center line due to improved nucleation. They also investigated the influence of four major input process parameters, namely beam size, spot-on time, pre-heat temperature and beam current, on the grain morphology. They showed via both numerical modeling and experimental investigation (using EBSD analysis) that the pre-heat temperature has the strongest impact on the grain morphologies. Yin et al. [77] developed a thermo-metallurgical model for studying the evolution of the grain growth angle during the L-PBF process of Ti6Al4V. They showed that the grain growth angle converges to a steady state value, after printing 4 – 5 layers, see Fig. 7(e). Manvatkar et al. [80] developed an FVM-based thermo-metallurgical model for DED of 316-L stainless steel. They found that the cooling rate decreases at higher layers, leading to larger dendritic arms spacing. It is worth to mention that apart from the mentioned spatially-resolved metallurgical models that solve the geometrical shape and size of individual grain structures, there is also another group of metallurgical models that predicts the phase fraction of different phases within the

Table 3
List of a number of volumetric heat sources used for MAM thermal modeling. OPD stands for optical penetration depth.

Name	Expression	Reference
Conico-Gaussian	$\frac{2\eta P}{OPD \cdot \pi r_b^2} e^{-\left(\frac{2(x^2+y^2)}{r_b^2}\right) \cdot (1-\frac{z}{H})}$	[66]
Cylindrical	$\frac{2\eta P}{OPD \cdot \pi r_b^2} e^{-\left(\frac{2(x^2+y^2)}{r_b^2}\right)}$	[60]
Decay-function	$\frac{2\eta P}{OPD \cdot \pi r_b^2} e^{-\left(\frac{2(x^2+y^2)}{r_b^2}\right) \cdot e^{-\left(\frac{z}{d}\right)}}$	[63]
Goldak	$\frac{6\sqrt{3}P\eta}{a_f bc\pi\sqrt{\pi}} e^{-\left(3(\frac{x^2}{a_f^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2})\right)} \quad \frac{6\sqrt{3}P\eta}{a_r bc\pi\sqrt{\pi}} e^{-\left(3(\frac{x^2}{a_r^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2})\right)}$	[64]
Gusarov	$\frac{Q_0\rho_h a}{(4\rho_h-3)D}\left([1-\rho^2]e^{-\lambda}[(1-a)e^{-2a\zeta}+(1+a)e^{2a\zeta}]-[3+\rho e^{-(2\lambda)}][(1+a-\rho_h(1-a))e^{2a(\lambda-\zeta)}+(1-a-\rho_h(1+a))e^{2a(\zeta-\lambda)}]\right) - \frac{3(1-\rho_h)(e^{-\zeta}-\rho_h e^{\zeta-2\lambda})}{4\rho_h-3}$	[65]
Vastola	$\frac{6f_2Q}{\pi r_c^2 h} e^{-\left(\frac{3(x^2+y^2)}{r_c^2}\right) \cdot (\frac{mz+r_v}{mh+2r_v})}$	[67]
Papadakis-Louizo	$\frac{3\eta P}{\pi H(r_e^2+r_e r_i+r_i^2)} e^{-\frac{(x^2+y^2)}{\left(r_e+\left(\frac{z}{H}\right)(r_i-r_e)\right)^2}}$	[68]
Bag-Trivedi-De	$\frac{9\eta P}{H \cdot \pi R^2(1-e^{-3})} e^{-\left(\frac{9(x^2+y^2)}{R^2\log[f_0](\frac{H}{z})}\right)}$	[69]

computational domain. These models are labeled as spatially-homogenized metallurgical models in literature [81–84]. Table 4 gives an overview of some of the important contributions to thermo-metallurgical simulations for MAM processes.

3.3. Thermo-mechanical models

As for the thermo-metallurgical models, meso-scale thermo-mechanical models typically also apply a pure conduction simulation to obtain the temperature field which is subsequently used as input for calculating thermal loads in the mechanical model. The governing equations for these models are summarized in Table 5 and by solving the static equilibrium equation using a proper material model, the displacement as well as stress/strain fields can be determined. Superscripts $()^{el}$, $()^{pl}$, $()^{tr}$, $()^{th}$ in Table 5 stand for elastic, plastic, phase transformation and thermal strains, respectively.

An example of such a model, is the work by Hussein et al. [88], who simulated the meso-scale thermo-mechanical behavior of 316-L steel during a 5-track L-PBF process and reported high magnitudes of von Mises stress due to cyclic cooling/heating. Gu and He [89] simulated the stress evolution during the L-PBF process of a Ni-Ti shape memory alloy. They found relatively higher residual stress magnitudes at the edges of the printed samples. Farahmand et al. [90] developed a thermo-mechanical model for DED of H13 and A36 steels and studied the impact of scanning velocity on the residual stresses. They reported higher magnitudes of transverse and longitudinal stresses at lower scanning speeds, where the melt pool was also found to be larger. Li et al. [91] modeled the thermo-mechanical conditions during a multi-layer L-PBF process and noted significant vertical residual stresses as more layers were built. Cheng, Shrestha and Chou [92] developed a meso-scale thermo-mechanical model to investigate the influence of scanning patterns on the magnitude of residual stresses during a 3-layer L-PBF process of IN718. They simulated the laser beam via a volumetric heat source that had a Gaussian planar profile and a parabolic distribution along the depth of the part.

Their simulations revealed that the 45-degree scanning pattern has the lowest residual stress magnitude in both longitudinal and normal directions, while the out-in pattern, led to the highest residual stresses. Parry et al. [93] studied the evolution of residual stresses in a single-layer multi-track L-PBF process for Ti6Al4V. They used Goldak's double-ellipsoid to simulate the laser beam's heating effect. They assumed bilinear-hardening for their material model and used temperature-dependent data for their simulations. Furthermore, they simulated irreversible powder-to-bulk transition in their model, assuming very low conductivity in the order of 0.1 W m^{-1} K^{-1} for the powder bed. Finally, they concluded that residual stresses parallel to the scanning pattern are much higher than the ones in the transverse direction, see Fig. 9(a).

As opposed to Parry et al. [93] and Cheng, Shrestha and Chou [92], who ignored the impact of phase transformation, Tan et al. [95] developed a more advanced thermo-metallurgical-mechanical model that incorporated the solid state phase transformation in a multi-layer L-PBF process of Ti6Al4V. Tan et al. [95] also assumed a bilinear-hardening material model and they noticed that accounting for solid state phase transformation causes higher compressive stresses locally, compared to a case where this effect is neglected, see Fig. 9(c). Chen et al. [96] studied the impact of different overlapping distances and scanning widths on the final residual stresses, see Fig. 9(d). They also ignored the influence of phase transformation in their work and finally recommended an overlapping distance of 25–50%, as it causes sufficient re-melting. Table 6 lists some of the most important meso-scale thermo-mechanical works in literature.

3.3.1. Capabilities and limitations of thermo-mechanical models

As previously mentioned, meso-scale thermo-mechanical models resolve the movement of the laser beam, and hence they can be used as a tool to study the effect of scanning pattern on residual stress. However, these models can only be applied to a small domain, involving few tracks and layers. It should be mentioned that a number of researchers such as Denlinger et al. [94] managed to simulate the thermo-elasto-plastic evolution during L-PBF for up to 38 layers for a cube, via an element coarsening strategy, see Fig. 9(b). At the moment, it is still not possible to model the deformation or stress distribution of a large part with dimensions in the order of centimeters, while at the same time being able to resolve the movement of the laser beam. Olleak and Xi [102] and Moran, Warner and Phan [103] managed to simulate the meso-scale melt pool movement in functional structural parts, using adaptive re-meshing and the principal of superposition, respectively. Nevertheless, they only simulated the thermal behavior and did not model the entire thermo-mechanical conditions for those functional parts. In this respect, multi-scaling can be pursued for part-scale thermo-mechanical modeling, which will be discussed in Section 5.

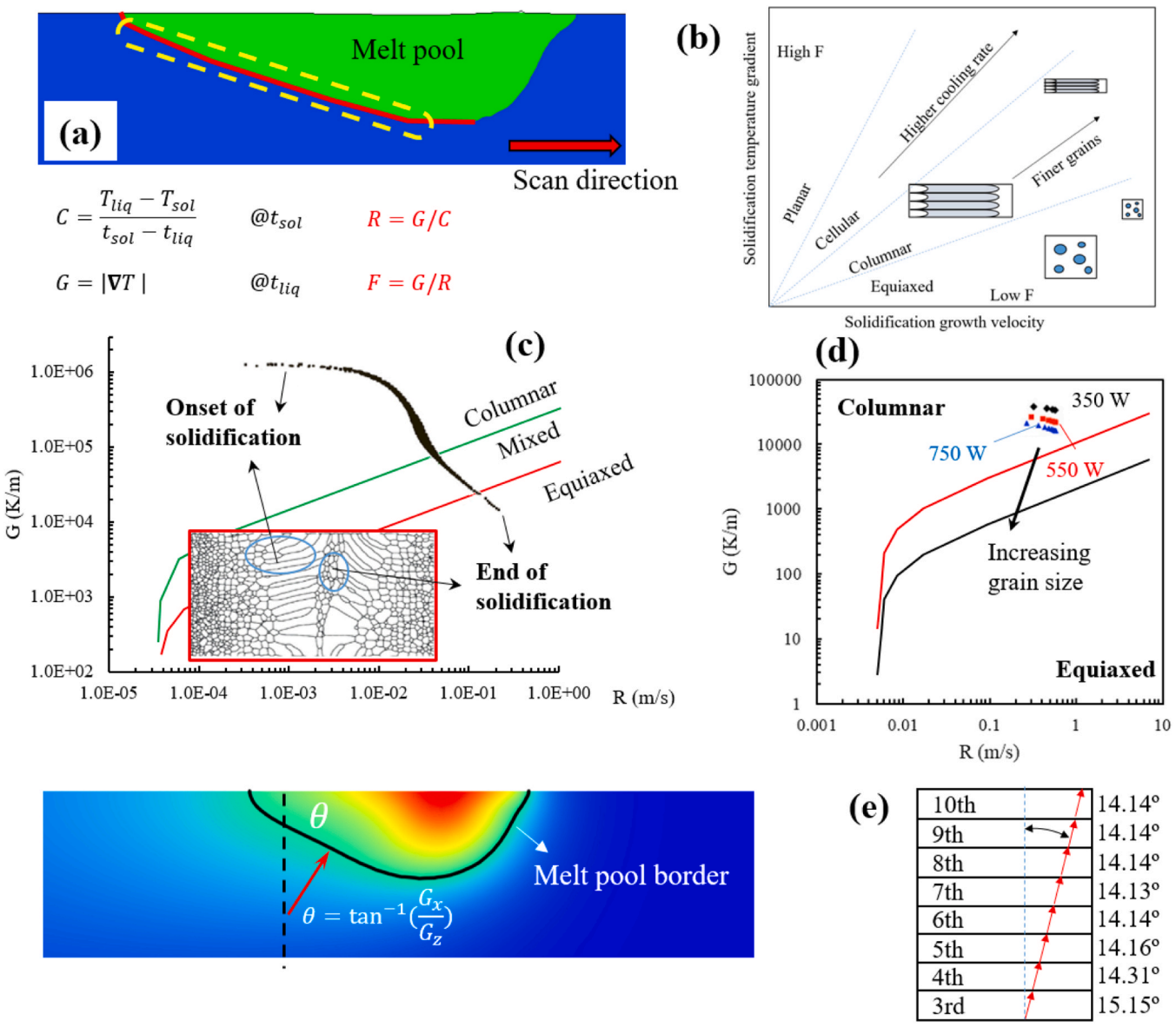


Fig. 8. (a) metallurgical variables at the melt pool border. (b) G - R curve schematically showing possible grain morphologies [73]. (c) calculated grain morphology by Raghavan et al. [74] (the simulated grain morphology is taken from [75]) and (d) Bontha et al. [76], for IN718 and Ti6Al4V, respectively. (e) predicted grain growth angle by Yin et al. [77].
All figures are reproduced from their original samples in their respective papers.

Table 4
An overview of some of the major thermo-metallurgical simulations of MAM.

Topic and highlights	Authors	Coupling	Software	Reference
Thermo-metallurgical modeling of MAM of Ti6Al4V	Gockel, Beuth, and Taminger 2014	Metallurgical	ABAQUS	[85]
Thermo-metallurgical modeling of MAM of IN718 + simulation of dendrite's growth and nucleation	Nie, Ojo, and Li 2014	Metallurgical /microstructural	ABAQUS	[86]
Investigation of the effect of involved process conditions on the metallurgical conditions of IN718 parts made with E-PBF	Raghavan et al. 2016	Metallurgical	Truchas code	[74]
Simulating the grain growth direction in multi-layer L-PBF of Ti6Al4V	Yin et al. 2018	Metallurgical	Coded in Matlab	[77]
Multilayer simulation of MAM of 316-L stainless steel + study the effect of heat accumulation on the dendrite sizes	Manvatkar et al. 2011	Metallurgical	FVM-based in-house code	[80]
Thermo-metallurgical modeling of MAM of Ti6Al4V using analytical solution	Bontha et al. 2009	Metallurgical	In-house code with Matlab	[76]
Thermo-metallurgical modeling of L-PBF of IN718 + parametric study on the role of process conditions on the columnar-to-equiaxed grain transition	Plotkowski, Kirka, and Babu 2017	Metallurgical	In-house code	[78]
Thermo-metallurgical modeling of DED of 316-L stainless steel using semi-analytical solution	Huang et al. 2019	Metallurgical	In-house code	[87]

Table 5
Governing partial differential equations for mechanical models.

Static equilibrium	$\frac{\partial \sigma_{ij}}{\partial x_j} = 0,$
Hooke's general law	$\sigma_{ij} = \frac{E}{1+\nu}\left[\frac{1}{2}\left(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}\right)+\frac{\nu}{1-2\nu}\delta_{ij}\delta_{kl}\right]\epsilon^{el}_{kl}$
Strain decomposition	$\dot{\epsilon}^{total}_{ij} = \dot{\epsilon}^{el}_{ij} + \dot{\epsilon}^{pl}_{ij} + \dot{\epsilon}^{th}_{ij} + \dot{\epsilon}^{tr}_{ij}$
J2 flow theory	$\dot{\epsilon}^{pl}_{ij} = \frac{9}{4}\left[\frac{1}{E_t}-\frac{1}{E}\right]\frac{s_{kl}\dot{\sigma}_{kl}}{\sigma^2_e}s_{ij}$
Strain-displacement relationship	$\varepsilon_{ij} = \frac{1}{2}\left[\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right]$

4. Meso-scale models: flow-based simulations

In the meso-scale flow-based simulations of MAM processes, as per their name, fluid-related variables such as pressure and velocity vectors are also calculated besides temperature, meaning that at least four more unknowns need to be solved for. This notably increases the required computational time and resources. CFD models for MAM processes can be subdivided into three general types, based on how they treat the free surface of the melt pool, i.e.: (1) assume a flat surface for the melt pool, (2) capture small deformations of the melt pool and (3) capture large melt pool deformations and porosities. The last one basically resolves the shape of every individual powder particle explicitly and is typically referred to as multi-physics simulation in MAM literature. These three types of simulations are discussed in the three following subsections. Table 7 lists the main governing equations and boundary conditions for CFD models of MAM processes. A_p (Pa) in Table 7 stands for the recoil pressure coefficient and γ (N m^{-1} K^{-1}) is the temperature sensitivity of surface tension.

4.1. CFD models with flat surface assumption

In this first type, neither recoil pressure nor surface tension effect are accounted for, as the free surface is flat and does not have any curvature that can lead to any surface tension forces. Nevertheless, the Marangoni effect is still taken into account in these models, since this is one of the main fluid dynamics forces in MAM and similar fusion-based processes [107–109]. It is worth mentioning that in this type of simulations, the powder bed is assumed to be a continuum with effective thermo-physical properties.

He, Fuerschbach and Debroy [110] developed a 3D CFD model for laser spot welding of 304 stainless steel. In their model, they assumed a flat surface for the melt pool and accounted for the Marangoni effect and buoyancy. They found, via a dimensionless analysis, that buoyancy plays a minimal role on the fluid flow inside the melt pool, compared to the Marangoni effect. Manvatkar, De and Debroy [111] developed a very similar model to their previous model in [110] and simulated multi-layer DED of 316-L stainless steel. As expected, they found that at the subsequent layers, cooling rate declines, leading to coarsening of the dendrites. Wei, Mazumder and Debroy [112] re-implemented this model to investigate the effect of different scanning directions on the metallurgical conditions during MAM of IN718. They found that unidirectional and back-and-forth scanning patterns cause the grains to grow with 60° and 45° angles with respect to the horizon, respectively. From the same research group, Mukherjee et al. [113] presented a modified version of the mentioned model and for the L-PBF process. Mukherjee et al. [114] then applied this model to investigate the effect of input parameters on metallurgical variables for four alloys. They noted that the cooling rate increases when the laser beam moves to a subsequent track, while this rate decreases with increasing number of layers. In another work, Mukherjee et al. [115] studied the formation of the lack-of-fusion porosities using this model. They came up with a dimensionless number that was used to give an indication of the formation of lack-of-fusion porosity. However, it must be underlined that lack-of-fusion porosities cannot be realistically described, solely by means of a flat-surface CFD model, since much of the porosity is essentially caused by powder denudation and the powder particles' overall size distribution. This necessitates development of a more complex model which includes the powder size distribution.

Zhang et al. [116] developed an FEM-based CFD model for the L-PBF process of IN718. They studied the relative strength of convection and conduction in heat transfer during the process and found that the convection heat flux is about 10 times higher than the one for conduction. Bayat et al. [66] modeled the thermo-fluid-metallurgical conditions during the L-PBF process of Ti6Al4V, using an FEM-based CFD model. They carried out a parametric study to analyze the impact of the scanning speed and beam size on the melt pool's thermo-fluid-metallurgical

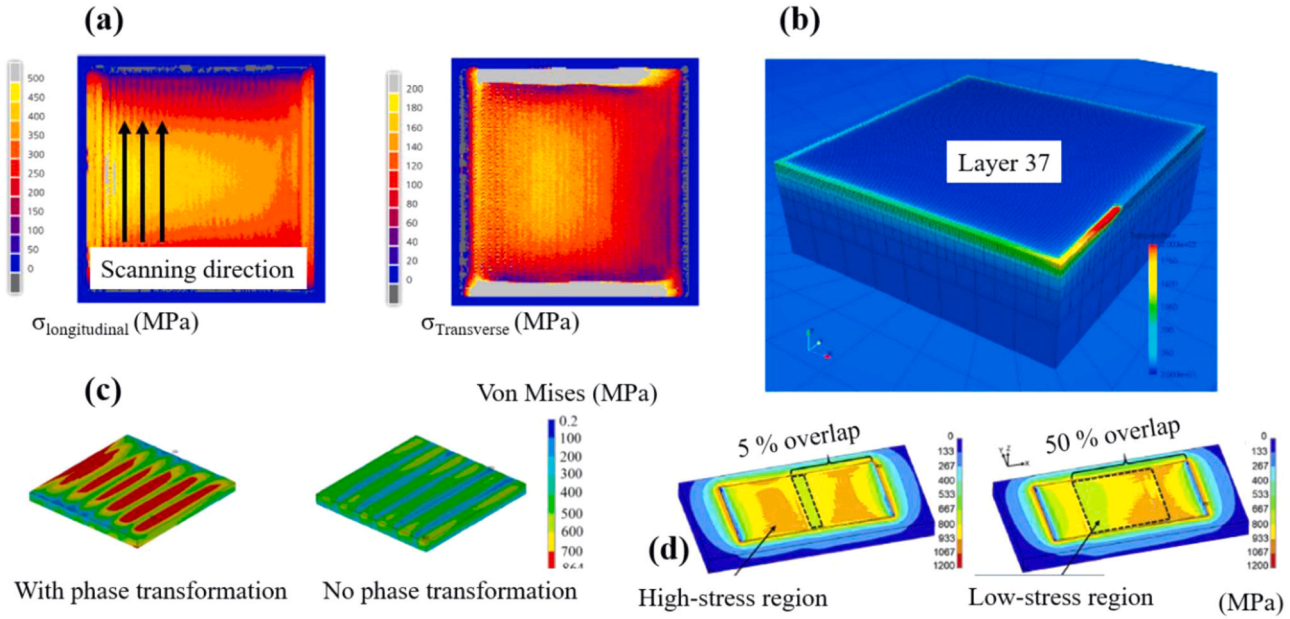


Fig. 9. (a) comparison between normal longitudinal and transverse residual stresses [93]. (b) temperature field at the 37th layer using an element coarsening method [94]. (c) comparison between equivalent residual stresses without and with the inclusion of phase transformation stresses [95]. The impact of overlapping distance of 5% and 50% on the final residual stresses [96].

Table 6
Highlights of a number of thermo-mechanical simulations for MAM.

Topic and highlights	Authors	Coupling	Software	Reference
Thermo-mechanical modeling of multi-layer L-PBF of 316-L stainless steel	Yingli Li et al. 2018	Mechanical	ANSYS	[91]
Thermo-mechanical modeling of 3-layer L-PBF of IN718 + study on the influence of different scanning patterns on the final residual stresses	Cheng, Shrestha, and Chou 2016	Mechanical	ABAQUS	[97]
Thermo-mechanical modeling of a single-layer multi-track L-PBF of Ti6Al4V	Parry, Ashcroft, and Wildman 2016	Mechanical	ABAQUS	[93]
Thermo-metallurgical-mechanical modeling of L-PBF of Ti6Al4V + accounting for solid state phase transformation	P. Tan et al. 2019	Mechanical/Metallurgical	ANSYS	[95]
Thermo-mechanical modeling of L-PBF of Ti6Al4V + study on the effect of different overlapping distances on the residual stress formation	C. Chen et al. 2019	Mechanical	ANSYS	[96]
Thermo-mechanical modeling of DED of 316-L stainless steel and mild steel + study the impact of process parameters on the residual stress formation	Farahmand and Kovacevic 2014	Mechanical	ANSYS	[90]
Thermo-mechanical modeling of 5-track L-PBF of 316-L stainless steel	Hussein et al. 2013	Mechanical	ANSYS	[88]
Thermo-mechanical modeling of L-PBF of AlSi10Mg + study on the impact of spot-on time on the residual stress magnitude	J. Wu, Wang, and An 2017	Mechanical	ABAQUS	[98]
Thermo-mechanical modeling of L-PBF of Ni-Ti shape memory alloy	Gu and He 2016	Mechanical	ANSYS	[89]
Thermo-elasto-plastic simulation of the L-PBF process of IN718 + simulation of 38 layers via an element coarsening strategy.	Denlinger et al. 2017	Mechanical	Pan Solver	[94]
Modeling of thermo-elsto-plastic conditions in overhang areas in L-PBF of 316-L	Hodge et al. 2014	Mechanical	Diablo	[99]
Thermo-mechanical modeling of L-PBF for 316-L stainless steel + sensitivity analysis for conductivity, specific heat capacity and Young's modulus	Conti et al. 2018	Mechanical	ANSYS	[100]
Thermo-mechanical modeling of the DED and L-PBF processes of IN718 + simulation of thin wall sample for DED and overhangs for L-PBF + validation w.r.t residual stress measurements from XRD	Song et al. 2020	Mechanical	ABAQUS AM module	[101]

conditions and found that, increasing the scanning speed leads to a significant rise in the solidification cooling rate, see Fig. 10(a).

Le and Lo [117] made an FVM-based CFD model to study the impact of different laser powers on the melt pool dimensions. They simulated the thermo-fluid conditions for a stainless steel with high sulfur content that leads to the inverse Marangoni effect, see Fig. 10(b). They noted that the model overestimates the melt pool depth in stable melt pool conditions, but underestimates it in the keyhole melt pool regime. The main reason for the overestimation is the fact that the recoil pressure, which leads to deep depressions, is not taken into account in their simulation. Jiang et al. [118] modeled the DED process of a Ni-based powder on Q235 steel with a flat surface assumption for the melt pool, although in reality, a bead is manufactured in the DED process, due to deposition of material. In their work, they found that the Marangoni effect leads to widening of the melt pool during the process. Table 8 lists some of the most important CFD works with flat surface assumption.

4.2. CFD models with small surface deformation

In this branch of CFD models, a surface-tracking technique is implemented that can predict small surface deformations. In this respect, as the free surface of the melt pool can have a curvature, the surface tension effect can be accounted for. Moreover, in these models, the recoil pressure can also be modeled. However, these surface-tracking techniques do not explicitly capture the shape of individual powder particles, especially for the L-PBF process, where a continuum approach is used instead. Most of the examples of this type of CFD models, can be found for the DED process, where the surface-tracking method predicts the bead shape, as the track is being manufactured.

Gan et al. [121] developed a CFD model to predict the melt pool shape and its fluid dynamics conditions during the DED process of a cobalt-based alloy on steel. They used a moving mesh method for tracking the free surface of the melt pool using the Arbitrary Lagrangian Eulerian (ALE) technique. They also defined two velocities for expressing the melt pool dynamics, an internal flow velocity and a moving boundary velocity due to the deposition of material. A snapshot of this model is shown in Fig. 11(a). In a subsequent work, Gan et al. [122] made a relatively similar model to simulate the melt pool evolution during DED of a nickel-based powder on cast iron. They approximated the free surface of the melt pool via solving an analytical force balance equation on the surface of the melt pool. They also reported larger dendrite sizes due to reduced cooling rates at the subsequent tracks. Gan et al. [123] re-implemented this model and applied it to study the effect of process parameters on the dilution and re-deposition of sulfur during DED. They showed that higher mass flow rate leads to sulfur dilution, which in turn causes an outward flow of liquid due to normal Marangoni flow. However, they showed that at lower mass flow rates, the sulfur concertation remains high in the melt pool, causing an inward flow due to inverse Marangoni flow. Kumar and Roy [124] developed an FVM-based CFD model of the DED process and expressed the shape of the melt pool's free surface using a dimensionless force balance equation on the surface. They also studied the impact of main input parameters on the thermo-fluid conditions of the melt pool via a dimensionless analysis. Zhao et al. [125] developed a CFD model of the DED process of high-speed steel on carbon steel, where they implemented a combined level-set volume of fluid method for predicting the free surface of the melt pool.

In most of the CFD simulations of the DED process, the addition of powder particles is simulated via a moving Gaussian-shape mass flux. The mass flux distribution, however, can be very different from the idealized Gaussian shape, if the nozzle is not perfectly focused on the build plate [128]. Some researchers managed to develop two separate models, one for simulating the powder injection and the other for simulating the melt pool thermal conditions [129,130]. However, it should be mentioned that Bayat et al. [126] managed to combine the two, such that the simulation of DED with explicit representation of

Table 7
List of governing equations for thermo-fluid CFD models implemented for MAM simulations [104–106].

Continuity	$\frac{\partial(\rho u_i)}{\partial x_i} = 0$	
Navier-Stokes	$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_k}{\partial x_k}\right)\right] - \frac{K_C(1-f_l)^2}{C_K + f_l^3}\bullet u_i$	
Surface-tracking	$\frac{\partial}{\partial t}(F) + \frac{\partial}{\partial x_j}(Fu_j) = 0$	
Interfacial forces	Recoil pressure	$P_{recoil} = A_p\left[P_{atm}\exp\left[f_{(i)}\right]\left(\frac{\Delta H_{lv}}{R\bullet T_{boil}}\left(1 - \frac{T_{boil}}{T}\right)\right)\right]$,
	The Marangoni effect	$\tau_{Marangoni} = \gamma[\nabla T - \mathbf{n}(\nabla T \bullet \mathbf{n})]$
	Surface tension effect	$P_{capillary} = (\sigma_0 + \gamma[T - T_l])\kappa$

powder particles was achieved, see Fig. 11(b). Furthermore, it must be noted that spattering of powder particles can affect the mass flow rate and this is usually compensated for via introducing a powder catchment efficiency in the simulations [131].

Chen et al. [132] developed an FEM-based model for simulating the thermo-fluid conditions during the L-PBF process of ceramic material. They used the level set method to track the free surface of the melt pool and similar to the models with flat free surfaces, they assumed a continuum layer with reduced material properties for modeling the powder layer. Queva et al. [127] modified this model and applied it to the L-PBF process of IN718 and Ti6Al4V, see Fig. 11(c). They also assumed a reduced-order continuum layer to represent the powder layer. The compressible Navier-Stokes equations were solved in the powder layer domain, taking the powder-to-liquid volume change of the layer into account. The same authors also found that neglecting evaporation can lead to wider and at the same time hotter melt pools. An overview of some of the most important contributions within this class of CFD models is given in Table 9.

4.3. Multi-physics CFD models

The surface-tracking algorithm implemented for this class of CFD models, enables explicit representation of individual powder particles and mainly for the L-PBF process. In this respect, as opposed to the previous models, the powder layer material properties will be identical to those of the bulk material. In this respect, this type of models is ideal for tracking lack-of-fusion and keyhole porosities forming during the process. Moreover, advanced laser-material interaction models such as ray-tracing, can be incorporated in these models as well.

Körner et al. [133] were among the first researchers in this field to develop a CFD model while being able to resolve the geometry of the individual powder particles. They used the Lattice Boltzmann Method (LBM) in their CFD simulations. In their work, they accounted for surface tension effects, wetting angles and phase transition and neglected other interfacial forces. Körner et al. [134] investigated the influence of a number of major input process parameters on the surface quality of E-PBF-made vertical walls. They mentioned that one can improve the quality of these walls via using a higher scanning speed and a smaller layer thickness. Bauereiß et al. [135] implemented this model to study the porosity formation during the E-PBF process. They investigated the impact of two input powers of 90 W and 180 W and noted that the former leads to channel-like defects that in their case started from layer 4 and continued all the way to layer 10. Markl et al. [136] went one step further and applied the 3D version of this model to the E-PBF process. They found that at a constant linear energy density, increasing the power and speed leads to deeper melt pools that in turn results in a significantly low porosity.

Khairallah and Anderson [137] developed an FEM-based CFD model for the L-PBF process of 316-L stainless steel using ALE3D (an Arbitrary Lagrangian Eulerian code developed at Lawrence Livermore National Laboratory). They studied the track uniformity at different scanning speeds and found out that at high scanning speeds, tracks will be discretized into separate spheres due to the Plateau-Rayleigh effect [138, 139]. Shrestha and Chou [140] developed an FVM-based CFD model for the L-PBF process of IN718 and in the same way as Khairallah and Anderson [137], neglected the recoil pressure and thermo-capillarity. Leitz et al. [141] also developed a similar model based on FEM and studied the track uniformity during the L-PBF process of molybdenum. Yu et al. [142] developed an FVM-based CFD model for L-PBF of aluminum and modeled the laser-material interaction via a moving volumetric heat source. They reported that both high and low linear energy densities can lead to rough surface finish, which is caused by the self-balling effect and lack of sintering. Xia et al. [143] studied the impact of linear energy density on the porosity formation during L-PBF of IN718. They developed an FVM-based model for their study and accounted for the recoil pressure and the Marangoni effect. A volumetric heat source model was used in their simulations for simulating the laser-heating effect. Furthermore, they reported that a scanning speed of 200 mm·s^{-1} would lead to a smooth surface finish, without any notable porosity, while a higher scanning speed of 500 mm s^{-1} results in open porosities on the surface of the sample. Nonetheless, it must be emphasized that in these works [142–144], the powder layer is assumed to be comprised of one layer of neatly-arranged single-sized powder particles, see Fig. 12(a). Whereas in reality, powder particles are randomly-stacked upon each other and have a relatively wide size distribution, see Fig. 12(b).

Khairallah et al. [148] developed a multi-physics CFD simulation of the L-PBF process of 316-L stainless steel using the ALE3D code discussed earlier. This work is basically a modification of the LLNL's previous work [137] and the recoil pressure and Marangoni effect were implemented in the model as well as an evaporation model. Lee and Zhang [149] developed a CFD model with comparable complexity as in [148] and simulated the melt pool's heat and fluid flow and metallurgical conditions based on a combined CFD-DEM simulation for L-PBF of IN718 and based on FVM. They also calculated the cooling rates to obtain the dendrite arm spacing during solidification. Wu et al. [150] developed an FVM-based simulation of the L-PBF process of H13 steel

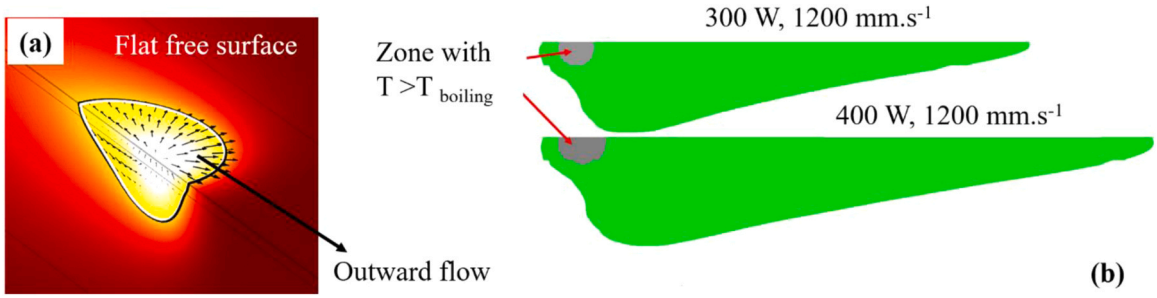


Fig. 10. Two examples of melt pool simulations with flat free surface: (a) Bayat et al. [66] and (b) Le and Lo. [117].

Table 8
List of CFD simulations for MAM with the flat surface assumption. FS stands for the flat surface assumption and Mar for Marangoni.

Topic and highlights	Authors	Physics	Software	Reference
3D thermo-fluid modeling of spot welding of 304 stainless steel using in-house FVM-based code	He, Fuerschbach, and DebRoy 2003	FS/ Mar	In-house code - FVM	[110]
3D thermo-fluid-metallurgical simulation of multi-layer DED of 316-L stainless steel + study the role of residual heat on the dendrite spacing	V. Manvatkar, De, and Debroy 2014	FS/ Mar	In-house code - FVM	[111]
Thermo-fluid-metallurgical modeling of grain growth evolution in MAM of IN718 with unidirectional and bi-derctional scanning directions	Wei, Mazumdar and Debroy 2015	FS/ Mar	In-house code - FVM	[112]
3D thermo-fluid modeling of L-PBF of 316-L, AlSi10Mg, Ti6Al4V and IN718 using OPD (optical penetration depth)-based volumetric heat source	Mukherjee et al. 2018a	FS/ Mar	In-house code - FVM	[113]
3D thermo-fluid-metallurgical modeling of L-PBF of 316-L, AlSi10Mg, Ti6Al4V and IN718 + study of the evolution of dendritic arm spacing in the subsequent tracks	Mukherjee et al. 2018b	FS/ Mar	In-house code - FVM	[114]
3D thermo-fluid modeling of L-PBF of 316-L, AlSi10Mg, Ti6Al4V and IN718 + study on the formation of the lack-of-fusion porosity	Mukherjee and DebRoy 2018	FS/ Mar	In-house code – FVM	[115]
3D thermo-fluid modeling of L-PBF of IN718 + comparison between the conduction and convection heat flux magnitudes within the melt pool	D. Zhang et al. 2018	FS/ Mar	COMSOL Multi-physics	[116]
3D thermo-fluid modeling of L-PBF of CL20ES powder + investigation of the impact of the process parameters on the melt pool shape	T. N. Le and Lo 2019	FS/ Mar	ANSYS Fluent	[117]
3D thermo-fluid modeling of DED of NiNbZrCo powder on Q235 mild steel	Jiang et al. 2020	FS/ Mar	ANSYS Fluent	[118]
3D thermo-fluid-metallurgical simulation of L-PBF of Ti6Al4V using FEM + conico-Gaussian volumetric heat source	Bayat et al. 2019	FS/ Mar	COMSOL Multi-physics	[66]
3D thermo-fluid-metallurgical simulation of L-PBF of Ti6Al4V using FEM + metallurgical-microstructural analysis	De Baere 2018	FS/ Mar	COMSOL Multi-physics	[119]
3D simulation of thermo-electric-magneto-hydrodynamic conditions for L-PBF of Al10Si	Kao et al. 2020	FS/ Mar	LBM	[120]

and studied the influence of evaporation and recoil pressure on the melt pool. They found that neglecting evaporation not only causes higher peak temperatures, but also leads to a wider melt pool.

Bayat et al. [145] looked into a different aspect of MAM and managed to simulate the lack-of-fusion porosity in a multi-track/multi-layer L-PBF process for IN718, using a combined CFD-DEM approach. They found that elongated lack-of-fusion porosities form at the two sides of the tracks, due to lack of proper heating from the laser beam at those locations. There are a number of modeling techniques that are typically used for simulating the powder-spreading step such as DEM [150], the rain drop method [135] and regular distribution [142]. As shown by Cao [151], the morphology of the powder layer can highly influence the track uniformity and shape. DEM is currently the most frequently-implemented method that accounts for mutual interaction between powder particles where usually Hertz and Voigt models are used for determining the normal and tangential forces between particles, respectively. Gu et al. [152] used a similar CFD-DEM technique as that of Bayat et al. [145] and simulated the multi-track/multi-layer melt pool behavior for a multi-material powder consisting of 316-L steel and Cu10Sn. Most of the CFD models used to simulate MAM concern metals/alloys that exhibit a normal (outward) Marangoni behavior. However, Bayat et al. [146] did a CFD simulation to understand the impact of different strengths of both normal and inverse Marangoni effects on the melt pools formed during L-PBF. They concluded that under the inverse Marangoni effect, the melt pool's width/depth ratio becomes small, see Fig. 12(d). They also found that regardless of the direction of the Marangoni effect, an increase in the strength of this interfacial force leads to a pronounced decrease in the melt pool temperature, mainly caused by improved heat transfer via convection. Bayat et al. [147] developed another FVM-based CFD model for simulating the keyhole formation during L-PBF, see Fig. 12(e). They showed that when the keyhole forms, a large number of laser rays will be entrapped in the keyhole depression walls, leading to multiple reflections that in turn significantly promote the heat transfer from the laser beam. They also simulated the porosity formation in a keyhole melt pool regime and validated their results with X-CT data from experiments. Tang, Tan and Wong [153] also studied the keyhole porosity formation for the L-PBF process with an FVM solver. They noticed that the previous track morphology can affect the track shape in the new layer.

Yan et al. [154] studied the track uniformity during E-PBF of Ti6Al4V using an FVM-based model. They suggested that a low layer thickness and a high laser power are beneficial for obtaining good track uniformity. In a later work, Yan et al. [155] implemented this model to investigate how different scanning patterns can lead to porosities. They found that the layer-wise interlace scanning pattern leads to low porosity during the E-PBF process. Tang, Le and Wong [156] modeled the humping formation during L-PBF of 316-L stainless steel. They showed via numerical simulation and experiments that, at a constant energy density, a higher laser power leads to more pronounced humps. Cao [151] used an FVM-based CFD model to investigate the impact of layer thickness, packing density and powder particles size distribution on track quality. Shi et al. [157] simulated the effect of different beam shapes on the heat and fluid flow and metallurgical conditions during L-PBF. They used two distinct beam shapes of longitudinal-elliptical and

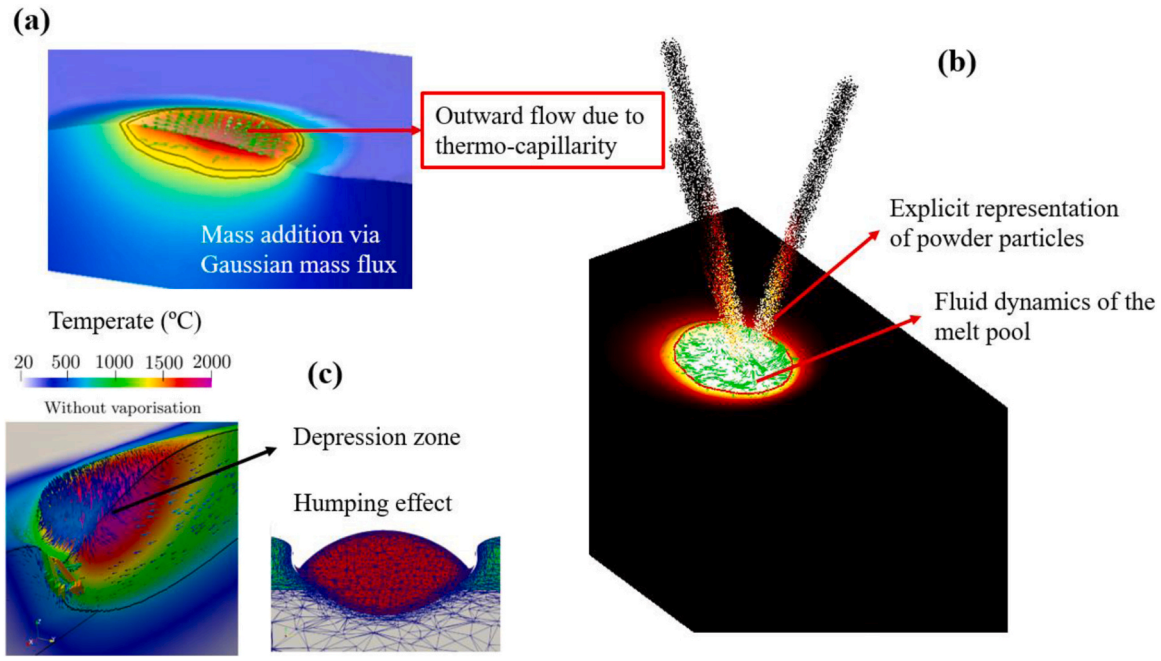


Fig. 11. (a) DED simulation using Gaussian mass flux as mass input by Gan et al. [121]. (b) CFD simulation of DED with explicit powder representation by Bayat et al. [126]. (c) the melt pool shape for L-PBF of IN718 without the inclusion of evaporation from Queva et al. [127].

Table 9
List of CFD simulations with small surface deformation assumption for MAM processes. SSD (small surface deformation), Mar (Marangoni), Cap (surface tension effects) and Recoil (recoil pressure).

Topic and highlights	Authors	Physics	Software	Reference
3D dimensionless thermo-fluid modeling of the DED process	Kumar and Roy 2009	SSD/ Mar, Cap	In-house code - FVM	[124]
3D thermo-fluid modeling of DED of nickel-based alloy on cast iron	Gan, Liu, et al. 2017	SSD/ Mar, Cap	In-house code	[122]
3D thermo-fluid-metallurgical modeling of DED of cobalt-based alloy on steel plate + analysis of dendritic arm spacing in the multi-layer process	Gan, Yu, et al. 2017a	SSD/ Mar, Cap	In-house code	[121]
3D thermo-fluid modeling of DED of sulfur-cobalt and sulfur-iron alloys + study the effect of different mass flow rates on the redistribution of sulfur	Gan, Yu, et al. 2017b	SSD/ Mar, Cap	COMSOL Multi-physics	[123]
3D thermo-fluid-metallurgical modeling of DED of T15 and T16/ CeO2 powders on low carbon plate using combined level set volume of fluid method	J. Zhao et al. 2020	SSD/ Mar, Cap	n. a.	[125]
3D thermo-fluid modeling of L-PBF of ceramic material using level set	Chen et al. 2018	SSD/ Mar, Cap	In-house code – FEM	[132]
3D thermo-fluid modeling of L-PBF of Ti6Al4V and IN718 using level set + study the impact of the recoil pressure on the melt pool	Queva et al. 2020	SSD/ Mar, Cap, Recoil	In-house code – FEM	[127]

transverse-elliptical and showed that the latter leads to a wide and shallow melt pool, while the former exhibits the opposite behavior. They also noticed that a longitudinal-elliptical beam leads to a columnar grain structure whereas the transverse-elliptical beam promotes nucleation of equiaxed grains. There is also a group of multi-physics models which does not require a surface-tracking scheme and instead relies on a Lagrangian description of the melt pool free surface. Smoothed Particles Hydrodynamics (SPH) is such a Lagrangian discretization method which is frequently used for predicting the free surface of the melt pool during MAM processes [158–161].

4.3.1. *Capabilities and limitations of multi-physics models*

It is well accepted that meso-scale CFD models are very useful for predicting detailed thermo-fluid behavior of the melt pool. As also shown earlier, these models are capable of predicting porosity formation and track the metallurgical evolution during the process. However, current multi-physics models have a number of limitations. For instance, the majority of these models neglect denudation. This phenomenon leads to dispersion of powder particles and consequently causes a non-homogenous surface finish in the as-built parts [162,163]. In the CFD models that are aimed at simulating high-energy L-PBF processes, this effect can be neglected, to some extent. However, for simulations addressing the final surface of a layer, neglecting this phenomenon can lead to large error. The second major issue for these models is their high computational expense. As shown earlier, the computational domains for these simulations are fairly small and usually cover a few tracks or layers. This can lead to severe overestimation or underestimation of the melt pool temperature, when neglecting the global location of the considered computation domain in the part. For instance, if the model is supposed to simulate the melt pool conditions at the last layers of a centimeter-size sample, the residual heat from the previous layers must be accounted for, either via an updated boundary condition or initial temperature of the domain. Table 10 gives an overview of some of the recent meso-scale CFD models for MAM that explicitly model the individual powder particles.

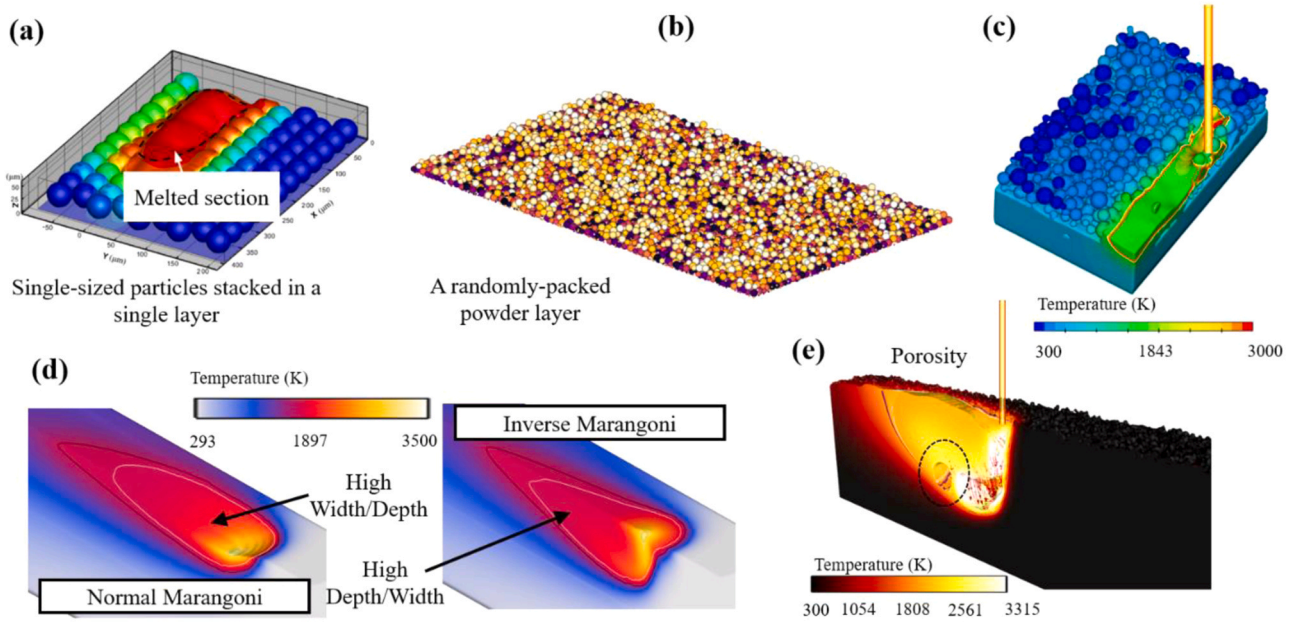


Fig. 12. (a) arranged single-sized powder layer CFD simulation from Xia et al. [144]. (b) randomly-packed powder layer from a DEM simulation. (c) multi-track-multi-layer simulation of the L-PBF process by Bayat et al. [145]. (d) comparison between the inverse and normal thermo-capillarity via CFD modeling [146]. (e) high-fidelity keyhole simulation of L-PBF of Ti6Al4V by Bayat et al. [147].

5. Part-scale models

As mentioned earlier, meso-scale simulations that capture the movement of the laser during the printing process, are standalone, quite time-consuming, as they need to solve for a large number of unknowns over a relatively small domain, potentially not covering more than a few tracks or layers. Furthermore, employing a meso-scale model for a real-size part would entail a sufficiently fine mesh size and a small time step, to fulfill the convergence criteria. This would then increase the total computational time that would itself require huge computational resources for running such jobs. In this respect, to lower the computational requirements and shorten the runtimes, different part-scale modeling techniques have been developed that allow for an accurate simulation of the thermo-mechanical conditions of a real-size part within an acceptable time. These methods are described in the two following subsections. Inherent strain-based methods are covered in the first subsection and other part-scale techniques are introduced in the second part.

5.1. Modified inherent strain method

The original inherent strain (IS) method was proposed in the 1970 s and was primarily applied to the welding field for residual distortion and stress prediction [177,178]. The key idea of the IS method is to replace the transient thermomechanical analysis involved in modeling the MAM process with a quasi-static mechanical analysis to reduce computational expense. This is made possible by modeling the local thermomechanical deformation induced by the heat source as the so-called “inherent strain”, which is applied to the quasi-static analysis model as initial strain. Considering that the MAM and conventional welding process have some common features (material deposition, melting and solidification, and intensive heat source), it is natural to apply the IS method to the MAM process. However, the original IS method shows poor performance when applied to the MAM process because the multi-track and multi-layer characteristics in MAM result in more complicated strain and stress evolution of the material. Therefore, different models and modifications have been proposed to improve the accuracy of the IS method. Most of the modifications focus on the calculation of inherent strains, while the implementation follows a layer-by-layer deposition manner.

In general, calculation of the inherent strain strains used in MAM can be classified into three methods: analytical, numerical, and empirical. The analytical method tries to find inherent strains from the material’s strain evolution during the MAM process. Yaghi et al. [179] used an isotropic compressive inherent strain vector to evaluate the distortion of a double cantilever beam fabricated by the L-PBF after cutting the support structure. The magnitude of the inherent strain is equal to the ratio of the yield strength to the Young’s modulus (at room temperature). Two FE models were studied when applying inherent strains. One has elastic material behavior, and the other has elastoplastic behavior. Although the elastic model was more computationally efficient than the elastoplastic model, the latter was able to provide more accurate distortion and residual stress predictions. The elastic model could roughly predict the distortion but overestimated the peak stress. Alvarez et al. [180] investigated the so-called inherent shrinkage method, where the thermal shrinkage strains are applied layer-by-layer in a elasto-plastic FE model to predict the distortion for the L-PBF process. The thermal shrinkage strain for each layer is equal to the material’s equivalent thermal expansion coefficient multiplied by the temperature difference between the melting point and the initial value of that layer. The equivalent thermal expansion coefficient is calculated based on the actual thermal expansion coefficient and is also related to the laser scanning strategy.

The numerical method usually involves hierarchical simulations where inherent strains are first determined in small representative models and then applied to simulate the actual part. Keller and Ploshikhin [181] developed a multi-scale simulation framework to achieve the fast prediction of distortion for MAM parts. The framework includes three models at different scales. First, the heat input is calibrated through a micro-scale heat source model. Then, the heat input is transferred into a meso-scale hatching model to calculate the inherent strains. Afterward, the inherent strains are applied to a macro-scale layer model to predict the distortion of the part. This framework has been experimentally validated using a stainless steel 316 L cantilever beam specimen. Liang et al. [182,183] and Chen et al. [184] developed a modified inherent strain (MIS) method for the DED and L-PBF processes,

Table 10
An overview of a number of recent meso-scale CFD simulations of MAM. LSD stands for large surface deformation. Mar, Cap and Recoil, stand for Marangoni, surface tension effects and the recoil pressure. LBM and SPH stand for Lattice Boltzmann and smoothed particle hydrodynamics.

Topic and highlights	Authors	Physics	Software	Reference
2D thermo-fluid modeling of multi-layer E-PBF of Ti6Al4V	Korner et al. 2011	LSD/, Cap	In-house code – LBM	[133]
Prediction of the quality of vertical walls made by E-PBF via LBM simulation	Korner et al. 2013	LSD/, Cap	In-house code – LBM	[134]
Simulation of intra-layer-track porosity formation during E-PBF of Ti6AL4V	Bauereiß et al. 2014	LSD/, Cap	In-house code – LBM	[135]
3D simulation of porosity formation and evolution during E-PBF of Ti6Al4V	Markl et al. 2015	LSD/, Cap	In-house code – LBM	[136]
3D thermo-fluid modeling of L-PBF of 316-L stainless steel + study on the balling effect and the Plateau-Rayleigh instability	Khairallah and Anderson 2014	LSD/Cap	ALE3D – FEM	[137]
3D thermo-fluid modeling of L-PBF of IN718	Shrestha and Chou 2018	LSD/Cap	ANSYS Fluent	[140]
3D thermo-fluid modeling of L-PBF of Molybdenum + study the impact of powder packing density on single tracks	Leitz et al. 2018	LSD/ -	COMSOL Multi-physics	[141]
3D thermo-fluid modeling of L-PBF of IN718 + investigation of the evolution of surface porosity	Xia et al. 2017 Xia et al. 2016 Yu, Shi, et al. 2016	LSD/Mar, Cap, Recoil	ANSYS Fluent	[143,144, 164]
3D thermo-fluid modeling of L-PBF of aluminum alloy + investigation of the evolution of the surface morphology	Yu, Gu, et al. 2016	LSD/Mar, Cap	ANSYS Fluent	[142]
3D thermo-fluid modeling of L-PBF of Ti6Al4V powder + comparison between the melting modes of conduction and keyhole conditions	Wang and Zou 2019	LSD/Mar, Cap	Flow-3D	[165]
3D multi-physics modeling of L-PBF of 316-L stainless steel + inclusion of a simplified version of multiple reflections	Khairallah et al. 2016	LSD/Mar, Cap, Recoil	ALE3D – FEM	[148]
3D multi-physics modeling of L-PBF of IN718 + determining the dendritic arm spacing via a metallurgical model + DEM simulation of the powder layer	Lee and Zhang 2016	LSD/Mar, Cap, Recoil	Flow-3D	[149]
3D multi-physics modeling of L-PBF of H13 steel + study on the impact of the evaporation on the melt pool conditions + DEM simulation of the powder layer	Y. C. Wu et al. 2018	LSD/Mar, Cap, Recoil	Flow-3D	[150]
3D multi-physics modeling of E-PBF of Ti6Al4V + study on the balling effect	Yan et al. 2017	LSD/Mar, Cap	Flow-3D	[154]
3D multi-physics modeling of multi-layer E-PBF of Ti6Al4V + study on the formation of lack-of-fusion porosities	Yan et al. 2018	LSD/Mar, Cap	Flow-3D	[155]
3D thermo-fluid modeling of L-PBF of 316-L stainless steel + comparison of keyhole and conduction regimes	Tan, Tang, and Wong 2018b Tan, Tang and Wong 2018 Tan, Tang, and Wong 2018a	LSD/Mar, Cap, Recoil	Open FOAM	[166–168]
3D multi-physics modeling of L-PBF of 316-L stainless steel + analysis of the keyhole regime and keyhole-induced porosities	Tang, Tan, and Wong 2018	LSD/Mar, Cap, Recoil	Open FOAM	[153]
3D modeling of the formation of lack-of-fusion porosity during multi-layer L-PBF of IN718	Bayat et al. 2019 a	LSD/Mar, Cap, Recoil	Flow-3D	[145]
3D modeling of the keyhole-porosity formation and evolution during L-PBF of Ti6Al4V	Bayat et al. 2019 b	LSD/Mar, Cap, Recoil	Flow-3D	[147]
3D CFD simulation of the impact of different strengths of normal and inverse Marangoni on the heat and fluid flow conditions of the melt pool	Bayat et al. 2021	LSD/Mar, Cap, Recoil	Flow-3D	[146]
3D multi-physics modeling of L-PBF of 316-L stainless steel + analysis of humping formation	Tang, Le, and Wong 2020	LSD/Mar, Cap, Recoil	Open FOAM	[156]
3D multi-physics modeling of L-PBF of 316-L stainless steel + analysis and comparison of different keyhole regimes	K. Q. Le, Tang, and Wong 2019	LSD/Mar, Cap, Recoil	Open FOAM	[169]
3D multi-physics modeling of L-PBF of 316-L stainless steel + study on the influence of powder packing density and size distribution on the tracks	Cao 2019	LSD/Mar, Cap, Recoil	Open FOAM	[151]
3D multi-physics modeling of beam-shaping during the L-PBF of 316-L using ALE3D	Shi et al.	LSD/Mar, Cap, Recoil	ALE3D – FEM	[157]
3D multi-physics modeling of spatter formation and surface porosity during L-PBF of Ti6Al4V	Qiu et al. 2015; Panwisawas et al. 2015	LSD/Mar, Cap, Recoil	Open FOAM	[170,171]
3D multi-physics modeling of melt pool stability during the L-PBF of 316-L stainless steel	W. Yuan et al. 2020	LSD/Mar, Cap, Recoil	ANSYS Fluent	[172]
3D multi-physics modeling of keyhole formation at different laser speeds during L-PBF of 304 stainless steel using level set	Kouraytem et al. 2019	LSD/Mar, Cap, Recoil	FEM-based code [105]	[173]
3D multi-physics modeling of defect formation and evolution during L-PBF of 316-L stainless steel	Zheng et al. 2019 a	LSD/Mar, Cap, Recoil	In-house code – LBM	[174]
3D multi-physics modeling of melt pool during pulsed L-PBF of 316-L stainless steel	Zheng et al. 2019 b	LSD/Mar, Cap, Recoil	In-house code – LBM	[175]
2D thermo-fluid model of L-PBF of 304 stainless steel	Shibai Liu et al. 2019	LSD/Mar, Cap	In-house code – SPH	[176]

see Fig. 13 for a schematic illustration. The MIS method includes two main steps: 1) a meso-scale detailed process simulation (based on a transient thermomechanical model using a moving pointwise heat source model) is employed to extract the inherent strain values through the MIS model from the strain history of the material; 2) a layer-by-layer quasi-static analysis using the inherent strains extracted in the previous step to predict residual stress and distortions for the entire part. This MIS method has been experimentally validated by thin-walled structures, double cantilever beams, and canonical parts made of Ti6Al4V and IN718, see Fig. 14. The method can provide more accurate predictions than the original IS method when applied to the MAM process. The MIS method has been applied to optimize the L-PBF process such as support structure design for crack prevention [35,185], support structure optimization [186], build orientation optimization [187], and scanning path optimization [188].

An overview of the most important contributions using the IS method is given in Table 11.

5.2. Multi-scaling methods

Flash heating (FH) is a part-scale multi-scaling technique and is primarily based on the layer-lumping concept. In this method, a part is divided into a certain number of blocks along the build direction. These blocks are essentially a number of actual layers which are merged together and denoted meta-layers. The meta-layer thickness is an input variable for the model and can vary. Upon the activation of these meta-layers, a volumetric heat source or a heat flux, starts heating the meta-layer during an on-time period. These heat sources depend on the input parameters of the process. Zaeh et al. [193] developed a part-scale model for simulating the residual stress formation during the L-PBF process of 1.2709 tool steel and used a meta-layer thickness of 1.00 mm. Prabhakar et al. [194] used this technique for modeling the thermo-mechanical conditions during E-PBF of IN718 coupons and used 700 μm thick meta-layers. They also noted that FH is unable to capture the impact of scanning patterns on the residual stresses.

The heat sources implemented for the FH method can vary depending on their on-time period. An on-time period of 1 – 10 ms leads to melting point temperatures during the activation of meta-layers [195]. However, if one extends the on-time period, to fulfill the total input energy, the magnitude of the heat source drops accordingly, as expected. This would lead to lower peak temperatures in the domain. For instance, Chiumenti et al. [196] used an extended on-time period for their pure thermal part-scale model. They reported peak temperatures that were far below the melting point of Ti6Al4V. Nevertheless, their results showed that the predicted temperature at the depth of the samples is in a good agreement with the data from thermocouples. It must be noted that such an extended on-time period does not lead to melting and does not cause sufficiently high temperature gradients to promote yielding. Thus, residual stresses do not form under such conditions.

Bayat et al. [197] studied the impact of the activation sequence on the thermal and mechanical conditions during L-PBF of Ti6Al4V. They compared the final deflection and residual stresses using 5 different meta-layer widths, ranging from 1 block per meta-layer to 20 blocks per meta-layer. They finally noticed that the FH method with 1 block per meta-layer overestimates the final deflection field and leads to unrealistic symmetrical deformation patterns. While the one with 20 blocks per meta-layer, led to a much better agreement between the experiments and simulations, see Fig. 15(a) and (b).

Another way of performing a part-scale simulation is to integrate the laser source in space and time. In such a technique, instead of modeling the thermal field by capturing the movement of the actual laser beam, the movement of an agglomerated heat source is modeled. Hodge et al. [198] developed a part-scale thermo-mechanical model and used a heat source that was 20 times bigger than the actual laser beam and at the same time had 20 time higher laser power. They applied this technique to several cases and predicted the final deformation and residual stresses in agreement with experimental results. Ganeriwala et al. [199] used this method to simulate the L-PBF of a bridge structure made of Ti6Al4V where the final strains were also in a very good agreement with the ones measured via XRD. One of the main drawbacks of the agglomerated heat

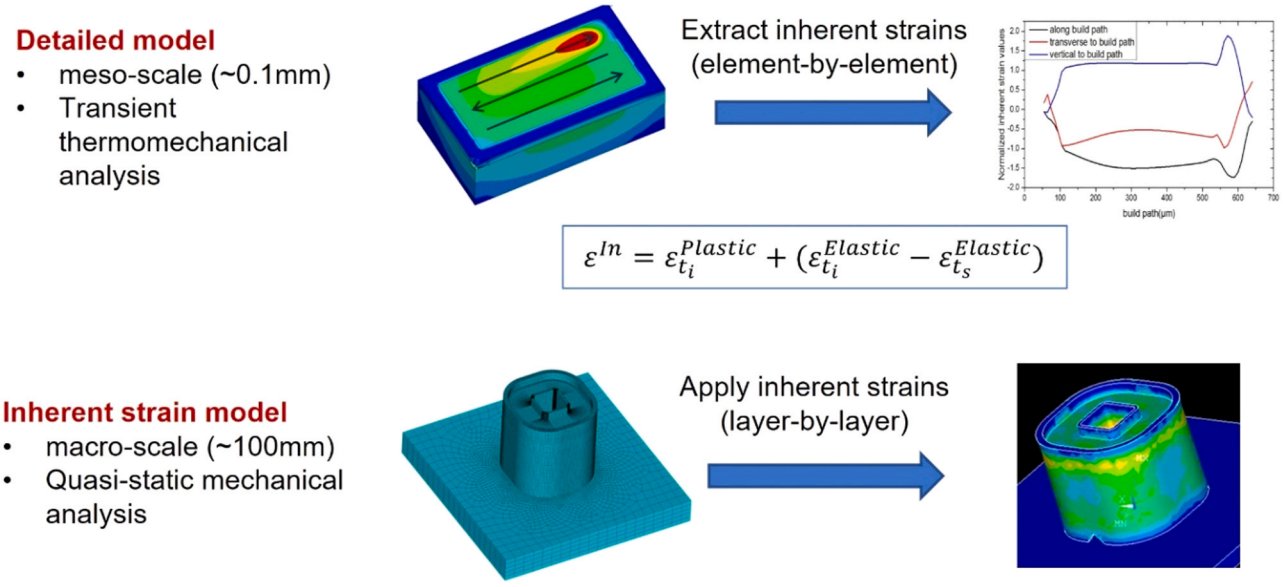


Fig. 13. Overview of the modified inherent strain method. In the equation, subscripts t_i and t_s are intermediate time and steady state time, respectively [182,183].

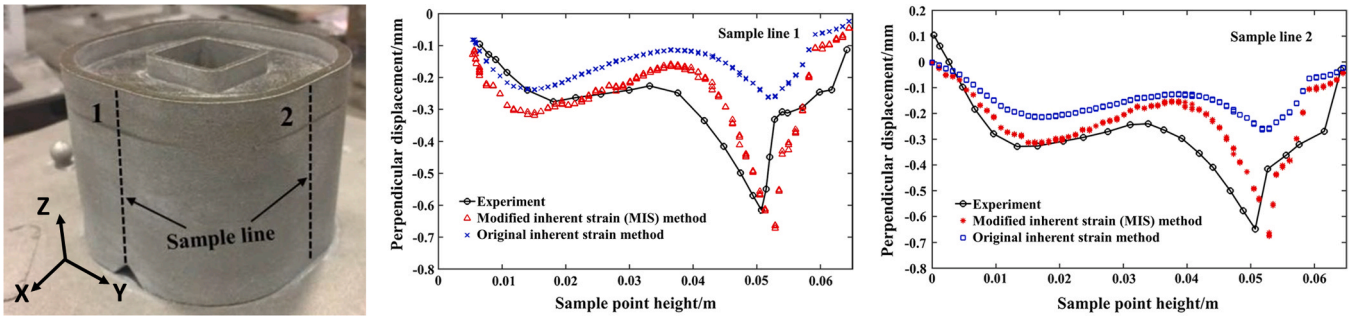


Fig. 14. Comparison of residual distortion between the modified inherent strain method and experimental measurement along two lines on a Ti6Al4V canonical part [189].

Table 11
Overview of models using the inherent strain method.

Topic and highlights	Authors	Implementation of IS	Reference
The original inherent strain method for predicting the distortion and residual stress for the welding process IS calculation: Experimental measurement	Ueda et al. 1975 Ueda et al. 1979	One-shot with elastic behavior	[177,178]
Comparison of residual stress and distortion prediction by analytical thermal model, elastic inherent model, and elastoplastic inherent model IS calculation: Yield strength over Young's modulus (at room temperature)	Yaghi et al. 2017	Layer-by-layer with elastic/ elastoplastic behavior	[179]
Evaluation of the residual stress and distortion of IN718 cantilever beams fabricated by SLM + study on the influence of scanning strategy IS calculation: Equivalent thermal expansion coefficient multiplied by the temperature difference between the melting point and the initial value of that layer	Alvarez et al. 2016	Layer-by-layer with elastoplastic behavior	[180]
Evaluation of the distortion of 316 L cantilever beams fabricated by SLM + study on distortion compensation IS calculation: Extracted from meso-scale FE simulation	Keller and Ploshikhin 2014	Layer-by-layer (the material behavior is unknown)	[181]
Evaluation of the residual stress and distortion for parts fabricated by DED and L-PBF with the modified inherent strain method IS calculation: Extracted from meso-scale FE simulation	Liang et al. 2018 Liang et al. 2019 Chen et al. 2019	Layer-by-layer with elastoplastic behavior	[182–184]
Calculation of inherent strains for an arbitrary hath pattern IS calculation: Neural network	Li and Anand 2020	Layer-by-layer with elastoplastic behavior	[190]
Evaluation of the distortion of maraging steel (18Ni300) double cantilever beams fabricated by L-PBF + study on the influence of scanning strategy + comparison with the original IS method IS calculation: Experimental calibration and optimization	Bugatti and Semeraro 2018	Layer-by-layer with elastoplastic behavior	[191]
Evaluation of the distortion of Ti6Al4V double cantilever beams fabricated by L-PBF + study on the influence of scanning strategy IS calculation: Experimental calibration and optimization	Setien et al. 2019	Layer-by-layer with elastic behavior	[192]

source (AG) method is its massive computational requirements, which is also underlined by Gouge et al. [200]. As an example, Hodge et al. [198] ran their AG thermo-mechanical model on systems with 128–512 CPUs, which are significantly stronger than most of nowadays typical desktop machines available. The other main challenge when using the AG method, is finding the optimal layer thickness and beam size agglomeration and this is still an active area of research. For instance, Hodge et al. [198] used agglomerated layers which were 20 times thicker than the actual layer thickness, whereas Ganeriwala et al. [199] used meta-layers with thicknesses 10 times thicker than actual layers. Furthermore, one other drawback of the AG method is its inability to model island scanning patterns where instead, a simplified serpentine pattern is typically used.

Rather than modeling the thermo-mechanical conditions in just one model, some researchers, however, did part-scale simulations via the development of a number of models at different scales involving different physics [201,202]. For instance, Li et al. [203] developed three models at three different scales for their part-scale deflection modeling. In their first model, they captured the movement of the laser beam in a meso-scale pure condition model. Then they developed a hatch-scale thermo-mechanical model where they used an equivalent heat flux that led to identical temperatures to those from the first model. Additionally, they applied the average residual stresses found from the hatch-scale model and finally mapped them to the final part-scale mechanical model, which was used to find the stresses and deformations.

Adaptive re-meshing has also shown its potential for performing part-scale simulations [204,205]. This is a unique technique which allows for resolving the actual laser beam movement at part scale. Olleak and Xi [102] developed a scan-wise adaptive re-meshing model to simulate the thermal conditions during the L-PBF process. In this method, they first meshed the sample with coarse elements and then they refined the mesh in the vicinity of the laser over a re-meshing area with a specific length. They modeled the melt pool movement during the printing of a bridge sample which had a size of 10 mm by 3 mm by 2 mm. Moran, Warner and Phan [206] developed a part-scale superposition-based FEM thermal model for the L-PBF process. They divided the temperature field into two fields of a nearly-analytical and subsequently corrected FEM temperature field, respectively. In their later work, the same authors [103], applied this technique to simulate the temperature field during the L-PBF process of an engine nacelle. Their model simulated the addition of 1313 layers and the actual G-code containing the laser patterns for each layer was taken into the model. They did the simulation in 18 h which was roughly less than half of the actual printing time for printing the component. They also defined two criteria for detecting the formation of lack-of-fusion and hot-spot defects, where they found that the latter increases steadily with the manufacturing of subsequent layers.

The meso-scale thermo-mechanical model developed by Cheng, Shrestha and Chou [92], took around 92 h to simulate the L-PBF process of a geometry with 6 mm by 6 mm by 90 μm size. While the part-scale FH thermo-mechanical model developed by Bayat et al. [197] took just less than 5 h for simulating the L-PBF process of a case-study with 30 mm by 30 mm by 3 mm. This once more highlights the computational gain when implementing multi-scaling techniques. An overview of some of the most important part-scale simulations are given in Table 12.

6. Conclusion

In the recent years and with the rapid growth of the AM industry, the numerical models of MAM processes have proven their capability as a cheap and reliable tool that can be utilized for predicting the quality of end-user products. These models can be used as a low-cost alternative to pure experimentation that would have otherwise slowed down the quality assurance and part optimization phases substantially and would have also incurred significant costs on the manufacturers.

MAM simulation covers a wide range both in terms of physics and length-scales. Based on the current literature, MAM simulations fall into three levels of length-scales; namely micro, meso and part-scale. Different modeling strategies as well as most important contributions

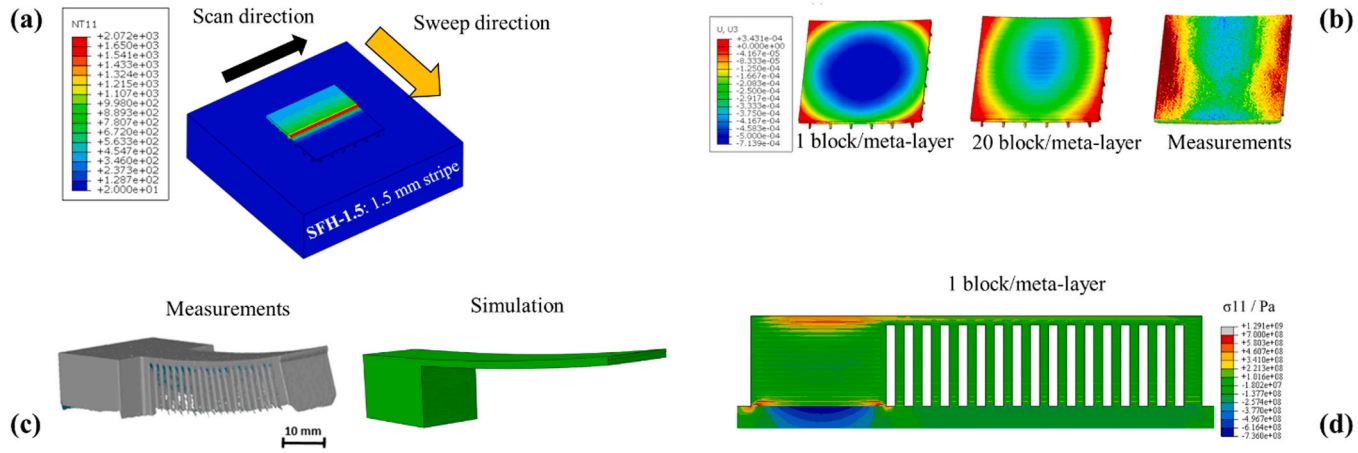


Fig. 15. (a) temperature field during a sequential flash heating simulation [197] and (b) the final deflection patterns with 1 and 20 block/meta-layers along with the experimental measurements [197]. (c) shows the comparison between the measured and predicted final deflection using the FH method with one block/meta-layer [46] and (d) shows the predicted residual stress right before the cutting step using FH with 1 block/meta-layer [46].

within this field belonging to meso-scale and part-scale simulations for both the DED and PBF processes are enumerated and discussed in this review. The meso-scale models are ideal for understanding the detailed melt pool behavior and they are subdivided into pure conduction models, thermo-metallurgical models, thermo-mechanical models and finally multi-physics thermo-fluid models. Pure conduction models are not very accurate in predicting the temperature fields within melt pools as they ignore a very important mechanism of heat transfer, the melt pool convection. However, this branch of meso-scale models is quite robust and they do not ask for significant computational resources. Therefore, the pure conduction models can be easily coupled to a subsequent mechanical or metallurgical model, where the former is used for predicting the formation of residual stresses and the latter is used for simulating possible phase transformations, respectively.

Multi-physics models, on the other hand, solve for more unknown variables, thus are much more complex compared to the pure conduction models. The multi-physics models also involve more sophisticated physics e.g. surface tension, the Marangoni effect, the recoil pressure, evaporation, etc. As a result, as these models cover more physical phenomena, they are very accurate in predicting the melt pool shape and size for both the PBF and DED processes. Depending on the degree of their complexity and the way they treat the free surface of melt pool, these models can also predict the formation of lack of fusion as well as keyhole-induced porosities in a reasonable agreement with experimental observations.

Although the multi-physics models are very accurate in predicting the melt pool morphologies, substantial effort is still needed for the calibration and validation of these models against experiments. This is to a large extent due to numerous uncertainties in either the process parameters or material properties. For instance, it is still quite challenging to measure the exact size of the laser beam for the PBF processes. Knowing the fact that the beam size is a first order process parameter, one should then expect noticeable over or underestimations of the melt pool shape and size in the models. The same applies for uncertainties in the value of the absorption coefficient, the surface tension forces or liquid viscosity at elevated temperatures as well. All these uncertainties would in turn lead to significant difficulties to validate these multi-physics models. The spatter and denudation phenomena also play a remarkable role and thus can highly impact the morphology of tracks made with the L-PBF process. Most of the current multi-physics models for the L-PBF process do not account for the spatter and denudation effects. Adding these physics would add extra unknowns to the problem and make the models even more time-consuming and complex. However, the current expectation is that the spatter and denudation effects will be an inherent part of the next generation of multi-physics models of MAM in the near future.

The main issue about the meso-scale models, as can also be inferred from their names, is that they are only applicable to a limited domain size, typically covering a few tracks or layers. Since these models capture the movement of the laser beam, they need to have a sufficiently fine mesh. This requirement is the main obstacle that hinders application of meso-scale models to a part-scale sample. Multi-scaling techniques are therefore a solution to this issue as they allow for the simulation of thermal or mechanical conditions at part-scale geometries within a reasonable timeframe of 10–100 h.

One of the potential investigation routes for the near future would be the involvement of reduced-order or surrogate models in the commercial printers' performance optimization feedback loop. At the moment, the majority of PBF printers use two sets of fixed process parameters; one for the bulk of the sample (raster) and one for contouring. This can, however, become problematic in case of an improper choice of input process parameters especially at critical locations such as overhang areas. Possible defects or process instabilities can show up in the form of dross formations and hotspots. One smart way to tackle these kind of geometry- or process-dependent defects would be via developing a closed-loop control system that adjusts and controls the process parameters in-situ and based on a well-calibrated surrogate model that itself is developed based on higher fidelity models.

The coupling between different length-scales is also an area of research within MAM that has not been explored sufficiently and this could furthermore be another potential investigation track for the upcoming years. For example, the material properties used for part-scale simulations are the ones from cast or wrought samples and these properties are quite different from those arising from samples made with MAM processes. One option would be to develop a micro-scale or meso-scale representative volume element that can both resemble the porosity and microstructural pattern of a sample made with MAM. Then a micro-mechanical analysis using crystal-plasticity can be carried out to find the nominal engineering mechanical properties that can later on be used in a more accurate part-scale simulation.

CRediT authorship contribution statement

Mohamad Bayat: Conceptualization, Investigation, Visualization, Writing – original draft. **Jesper Thorborg:** Conceptualization, Writing – review & editing. **Albert C. To:** Conceptualization, Investigation, Visualization, Writing – original draft. **Jesper Henri Hattel:** Conceptualization, Supervision, Funding acquisition, Resources, Writing – review & editing.

Table 12
An overview of a number of part-scale thermo-mechanical models using different multi-scaling methods. AG (agglomerated heat source), MS (multi-step method), FH, SFH (flash heating and sequential flash heating), AMR (adaptive mesh refinement).

Topic and highlights	Authors	Method	Software	Reference
Part-scale thermo-mechanical modeling of L-PBF of 316-L samples + experimental validation of vertical and horizontal prisms and an L-shape sample	Hodge et al. 2016	AG	Diablo [207]	[198]
Part-scale thermo-mechanical modeling of L-PBF of Ti6Al4V sample + analysis of rate-independent, viscoplastic and viscoelastic material models	Ganeriwala et al. 2019	AG	Diablo [207]	[199]
Part-scale simulation of thermal and mechanical conditions via development of micro, meso and macro-scale models	Li et al. 2017 Li et al. 2018 Li et al. 2016	MS	ABAQUS	[201–203]
Part-scale thermo-mechanical modeling of industrial components made of IN625, IN718 and AlSi10Mg + adaptive remeshing technique	Gouge et al. 2019	MS	Pan Solver (Autodesk)	[200]
Thermo-mechanical modeling of L-PBF of cantilever beam made of 1.2709 tool steel	Zaeh et al. 2010	FH	ANSYS	[193]
Part-scale thermo-mechanical modeling of E-PBF of coupons made of IN718	Prabhakar et al. 2015	FH	ABAQUS	[194]
2D thermo-mechanical modeling of L-PBF of Ti6Al4V samples + study on the impact of different flash-heating times on the stress field	Zhang et al. 2019	FH	ABAQUS	[195]
Part-scale thermo-mechanical modeling of DED of S-shape and rectangular samples made of Ti6Al4V	Lu et al. 2019	FH	Comet [208]	[209]
Part-scale thermo-mechanical modeling of L-PBF of IN718 and stainless steel 316-L + activation of layers with predefined initial melt temperature	Williams et al. 2018	FH	ABAQUS	[210]
Part-scale thermo-mechanical modeling of the L-PBF process of Ti6Al4V plate samples with sequential flash heating + comparison between different meta-layer activation strategies for the final deflections	Bayat et al. 2020	FH/SFH	ABAQUS	[197]
Part-scale thermo-mechanical simulation of L-PBF of a cantilever beam made of Ti6Al4V using FH + modeling of the heat-treatment stress relief step	De Baere et al. 2021	FH	ABAQUS	[46]
Part-scale thermo-mechanical modeling of DED of 316 SS using layer-by-layer FH for a single track and a multi-layer-track printing of a 35 mm^3 sample	Kiran et al. 2020	FH/SFH	ABAQUS	[211]
Part-scale thermal modeling of the L-PBF process using a scan-wise adaptive re-meshing technique	Olleak 2020	AMR	ANSYS	[102]
Part-scale superposition-based FEM thermal modeling of the L-PBF process of Ti6Al4V + modeling the printing of 1313 layers and simulating the actual laser movement according to the G-code	Moran et al. 2018 Moran et al. 2021	AMR	In-house code	[103,206]

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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